

CREDIT RISK PYTHON ANALYSIS

September 22, 2026

```
[2]: # DATA LOADING AND INSPECTION
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
data = pd.read_csv(r'C:\Users\User\Downloads\DATA ANALYSIS PORTFOLIO\5 WATER_
↳PROJECTS\SACCO LOAN DEFAULT PREDICTION\kenya_sacco_credit_risk_portfolio.
↳csv')
```

```
[3]: data.head()
# the first five rows
```

```
[3]:  loan_id  borrower_id  application_date      branch      county  age  \
0  5009570      109570      2024-01-01  Nairobi CBD      Nairobi   21
1  5046553      146553      2024-01-01        Nyeri  Kakamega   45
2  5000353      100353      2024-01-01      Kisumu      Nairobi   50
3  5013827      113827      2024-01-01        Meru      Kisumu   27
4  5008668      108668      2024-01-01  Westlands  Uasin Gishu   53

   employment_type  monthly_income_kes  membership_years  savings_balance_kes  \
0          Casual          28600          2          44900
1          Salaried          50400          5          91400
2  Business Owner          64200          8         174700
3  Self-Employed          52900          1         223200
4          Salaried         117000          2         309300

   ...  mobile_money_active_months  utility_payment_consistency  \
0  ...          22          0.753
1  ...          53          0.754
2  ...          93          0.777
3  ...          11          0.747
4  ...          21          1.000

   income_consistency_score  days_overdue  late_payment_count_current  \
0          0.642          6          0
1          0.940          3          0
2          0.938          1          0
3          0.735          3          0
4          1.000          4          3
```

	repayment_ratio	loan_status	default	generated_default_probability \
0	0.873	Late	0	0.0260
1	0.965	Current	0	0.0649
2	0.833	Current	0	0.0229
3	0.882	Late	0	0.0715
4	1.018	Late	0	0.0487

	risk_category
0	Low
1	Low
2	Low
3	Low
4	Low

[5 rows x 32 columns]

```
[4]: data.shape
# 50000 rows and 32 columns
```

```
[4]: (50000, 32)
```

```
[5]: # INSPECT THE TARGET VARIABLE
# our target variable = default( 0 = non-default and 1 = default)
data['default'].value_counts()

# The number of non-defaults(0) = 45217
# The number of defaults(1) = 4783
```

```
[5]: default
0    45217
1     4783
Name: count, dtype: int64
```

```
[6]: data['default'].value_counts(normalize = True) * 100

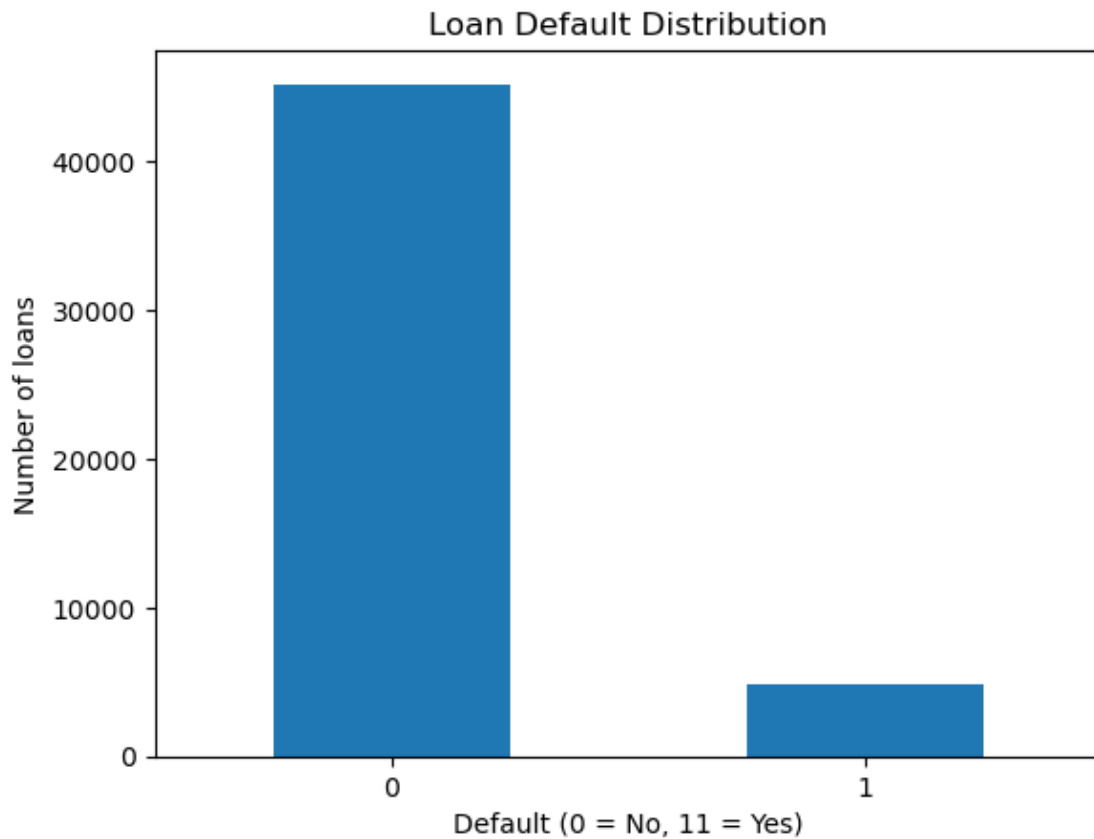
# default rate = 9.6%
# non-default rate = 90.43%

# remember, there are 50000 loans(100%) and default rate is roughly 10%(5000 ↵
↳ loans).
# the ratio of defaulted loans to total loans(5000:50000) is 1:10
# this means in every 10 loans, 1 loan will be defaulted
```

```
[6]: default
0    90.434
1     9.566
```

Name: proportion, dtype: float64

```
[7]: # visualize default distribution
data['default'].value_counts().plot(kind='bar')
plt.title('Loan Default Distribution')
plt.xlabel('Default (0 = No, 11 = Yes)')
plt.ylabel('Number of loans')
plt.xticks(rotation=0)
plt.show()
```



```
[8]: # CREATE A CLEANER DEFAULT RATE TABLE
default_summary = (
    data['default']
    .value_counts()
    .rename(index={0: 'Non-Default', 1: 'Default'})
    .to_frame('Number_of_Loans')
)

default_summary['Percentage'] = (
    default_summary['Number_of_Loans'] /
```

```

    default_summary['Number_of_Loans'].sum() * 100
)

default_summary

```

```

[8]:
      Number_of_Loans  Percentage
default
Non-Default      45217      90.434
Default          4783       9.566

```

```

[9]: # Business interpretation
# For the SACCO, the 9.57% default rate becomes our baseline.
# Later, when we build the ML model, we won't simply ask: "How accurate is the
    ↪ model?"
# We'll ask more important questions such as: How well can the model identify
    ↪ the 9.57% of borrowers who default?
# That's where precision, recall, F1-score, ROC-AUC, and the confusion matrix
    ↪ become important.

```

```

[10]: # BIVARIATE ANALYSIS

# NUMBER OF DEFAULTS BY LOAN PRODUCT
# we use groupby() function when checking the relationship/comparison between
    ↪ two variables
data.groupby('loan_product')['default'].agg(['count', 'sum', 'mean']).
    ↪ sort_values('mean', ascending=False)

```

```

[10]:
      count  sum  mean
loan_product
School Fees   7048   950  0.134790
Emergency     7846  1032  0.131532
Agriculture   7113   674  0.094756
Business      9972   851  0.085339
Development  11989   879  0.073317
Asset Finance  6032   397  0.065816

```

```

[11]: # DEFAULT RATE BY LOAN PRODUCT

loan_products_default_count = (
    data.groupby('loan_product')['default']
    .agg(['count', 'sum', 'mean'])
)

loan_products_default_count['default_rate_%'] =
    ↪ loan_products_default_count['mean']*100

```

```
loan_products_default_count.sort_values('default_rate_', ascending=False)
```

```
[11]:
```

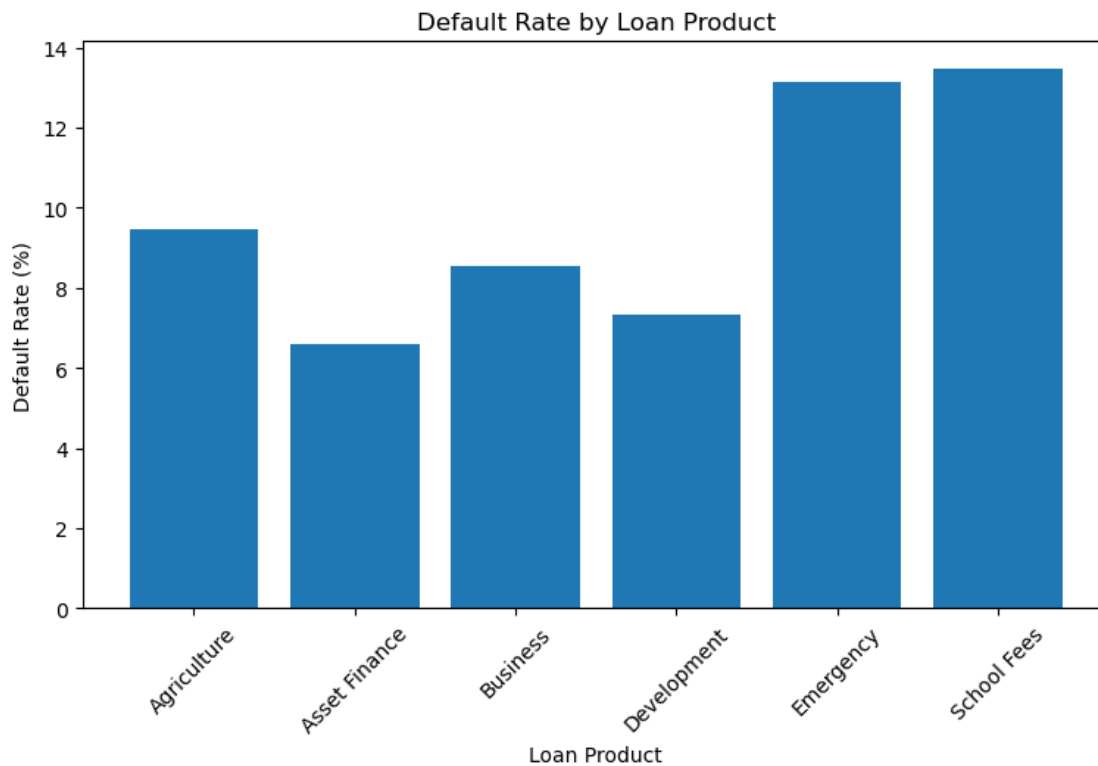
	count	sum	mean	default_rate_%
loan_product				
School Fees	7048	950	0.134790	13.479001
Emergency	7846	1032	0.131532	13.153199
Agriculture	7113	674	0.094756	9.475608
Business	9972	851	0.085339	8.533895
Development	11989	879	0.073317	7.331721
Asset Finance	6032	397	0.065816	6.581565

```
[12]: plt.figure(figsize=(9,5))

plt.bar(
    loan_products_default_count.index,
    loan_products_default_count['default_rate_%']
)

plt.title('Default Rate by Loan Product')
plt.xlabel('Loan Product')
plt.ylabel('Default Rate (%)')
plt.xticks(rotation=45)

plt.show()
```



```
[13]: # NUMBER OF DEFAULTS BY EMPLOYMENT TYPE
data.groupby('employment_type')['default'].agg(['count', 'sum', 'mean']).
    ↪sort_values('mean', ascending=False)
```

```
[13]:
```

	count	sum	mean
employment_type			
Casual	4424	567	0.128165
Farmer	5979	606	0.101355
Business Owner	9027	843	0.093387
Self-Employed	13479	1245	0.092366
Salaried	17091	1522	0.089053

```
[14]: # DEFAULT RATE BY EMPLOYMENT TYPE
employment_type_default_count = (
    data.groupby('employment_type')['default']
    .agg(['count', 'sum', 'mean'])
)

employment_type_default_count['default_rate_%'] =
    ↪employment_type_default_count['mean']*100

employment_type_default_count.sort_values('default_rate_%', ascending=False)
```

```
[14]:
```

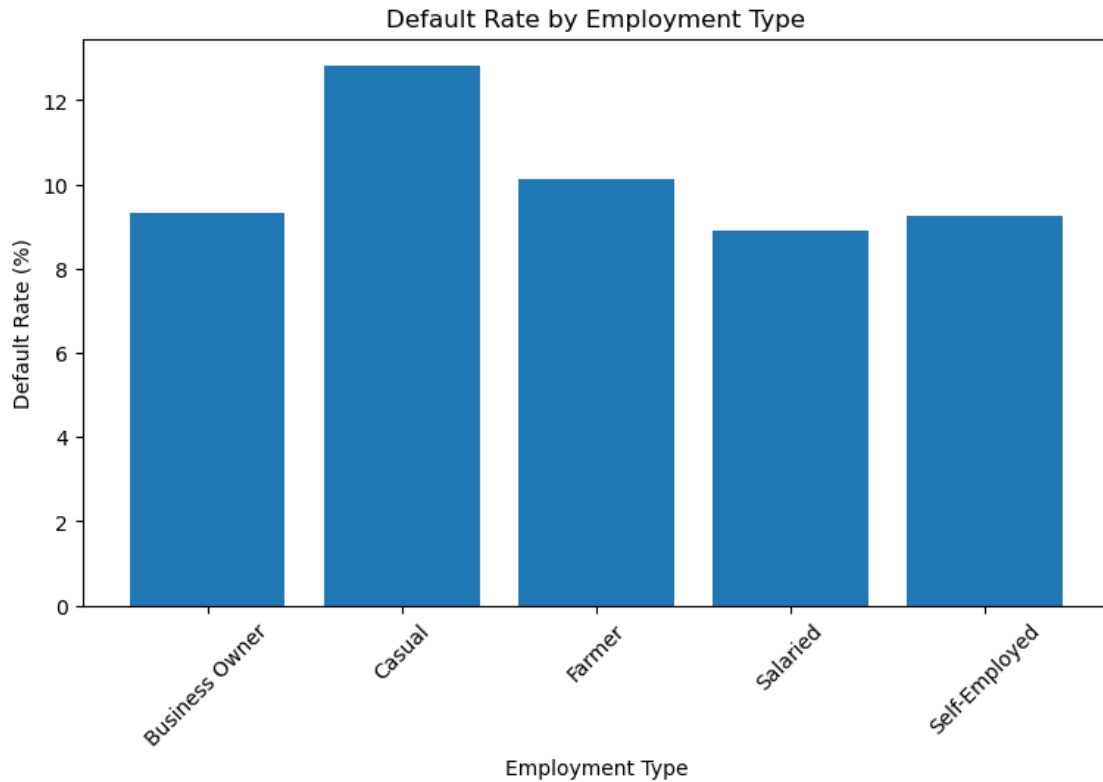
	count	sum	mean	default_rate_%
employment_type				
Casual	4424	567	0.128165	12.816456
Farmer	5979	606	0.101355	10.135474
Business Owner	9027	843	0.093387	9.338651
Self-Employed	13479	1245	0.092366	9.236590
Salaried	17091	1522	0.089053	8.905272

```
[15]: plt.figure(figsize=(9,5))

plt.bar(
    employment_type_default_count.index,
    employment_type_default_count['default_rate_%']
)

plt.title('Default Rate by Employment Type')
plt.xlabel('Employment Type')
plt.ylabel('Default Rate (%)')
plt.xticks(rotation=45)

plt.show()
```



```
[16]: # LOAN AMOUNT VS DEFAULT
data.groupby('default')['loan_amount_kes'].agg(
    ['count', 'mean', 'median', 'std', 'min', 'max']
)
```

```
[16]:
```

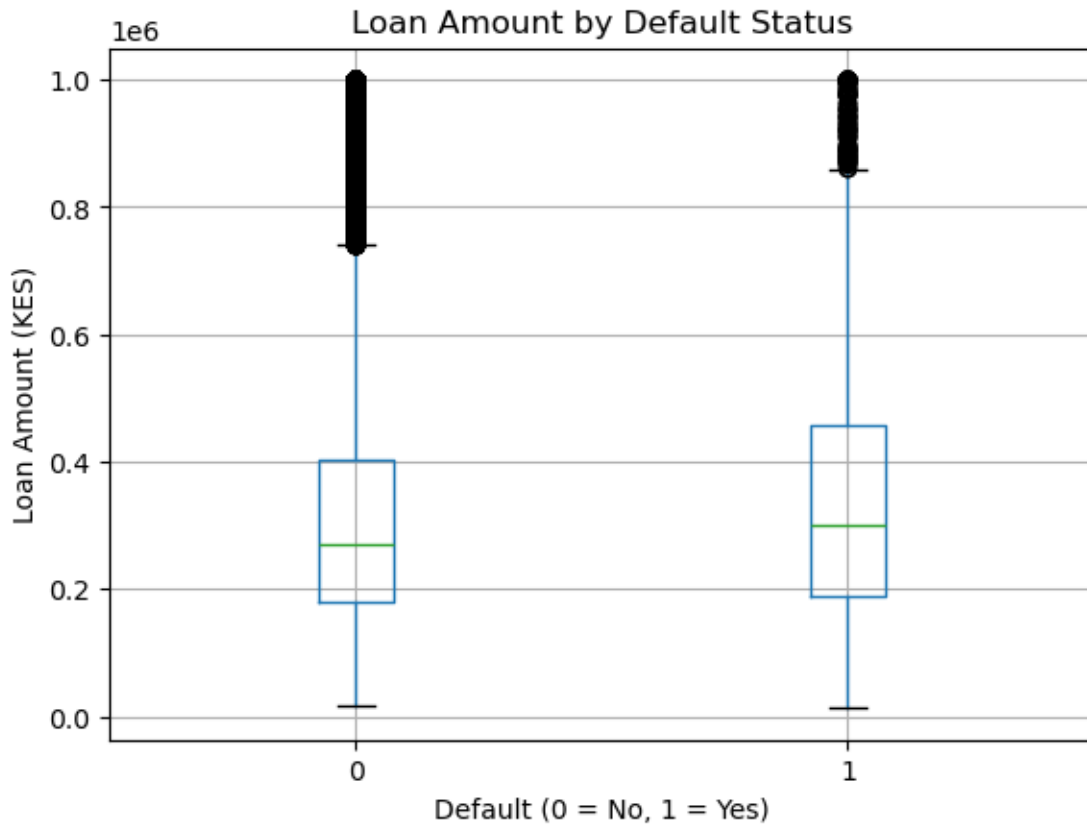
	count	mean	median	std	min	max
default						
0	45217	313542.813544	271500.0	185200.584546	16600	1000000
1	4783	345318.921179	299900.0	211601.990875	13000	1000000

```
[17]: #plt.figure(figsize=(8,5))

data.boxplot(
    column='loan_amount_kes',
    by='default'
)

plt.title('Loan Amount by Default Status')
plt.suptitle('')
plt.xlabel('Default (0 = No, 1 = Yes)')
plt.ylabel('Loan Amount (KES)')
```

```
plt.show()
```



```
[18]: # NUMBER OF DEFAULTS BY COUNTY
data.groupby('county')['default'].agg(['count', 'sum', 'mean']).
    sort_values('mean', ascending=False)
```

```
[18]:
```

	count	sum	mean
county			
Nyeri	2008	203	0.101096
Kakamega	2582	260	0.100697
Kiambu	5060	507	0.100198
Machakos	3333	327	0.098110
Nakuru	4563	445	0.097524
Kisumu	3033	295	0.097263
Uasin Gishu	3061	297	0.097027
Meru	3467	335	0.096625
Embu	2516	234	0.093005
Mombasa	3519	326	0.092640
Nairobi	12498	1156	0.092495
Other	4360	398	0.091284

```
[19]: # DEFAULT RATE BY COUNTY
county_default_count = (
    data.groupby('county')['default']
        .agg(['count', 'sum', 'mean'])
)

county_default_count['default_rate_%'] = (
    county_default_count['mean'] * 100
)

county_default_count.sort_values(
    'default_rate_%',
    ascending=False
)
```

```
[19]:
```

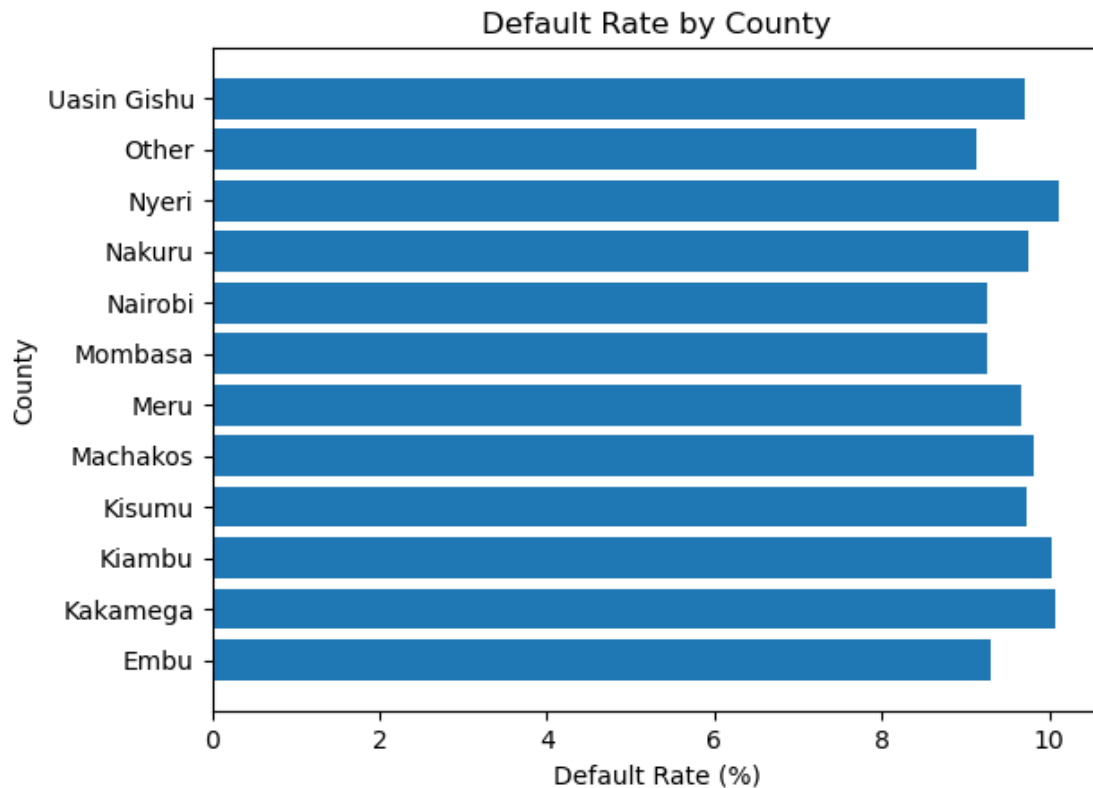
	count	sum	mean	default_rate_%
county				
Nyeri	2008	203	0.101096	10.109562
Kakamega	2582	260	0.100697	10.069713
Kiambu	5060	507	0.100198	10.019763
Machakos	3333	327	0.098110	9.810981
Nakuru	4563	445	0.097524	9.752356
Kisumu	3033	295	0.097263	9.726344
Uasin Gishu	3061	297	0.097027	9.702712
Meru	3467	335	0.096625	9.662532
Embu	2516	234	0.093005	9.300477
Mombasa	3519	326	0.092640	9.263995
Nairobi	12498	1156	0.092495	9.249480
Other	4360	398	0.091284	9.128440

```
[20]: #plt.figure(figsize=(10,8))

plt.barh(
    county_default_count.index,
    county_default_count['default_rate_%']
)

plt.title('Default Rate by County')
plt.xlabel('Default Rate (%)')
plt.ylabel('County')

plt.show()
```



```
[21]: #INCOME VS DEFAULT
data.groupby('default')['monthly_income_kes'].agg(
    ['count', 'mean', 'median', 'std', 'min', 'max']
)
```

```
[21]:
```

	count	mean	median	std	min	max
default						
0	45217	64734.827609	59500.0	30658.453057	10000	365400
1	4783	63923.938950	58300.0	31479.322275	10000	322100

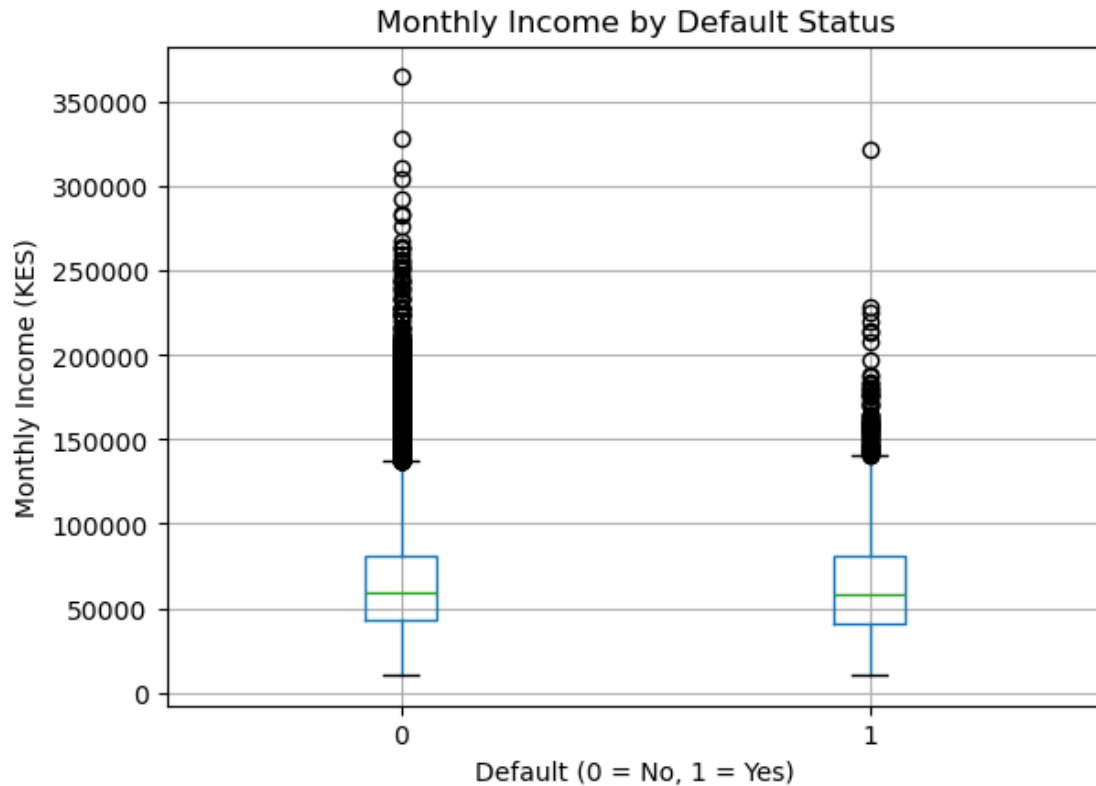
```
[22]: plt.figure(figsize=(8,5))

data.boxplot(
    column='monthly_income_kes',
    by='default'
)

plt.title('Monthly Income by Default Status')
plt.suptitle('')
plt.xlabel('Default (0 = No, 1 = Yes)')
plt.ylabel('Monthly Income (KES)')
```

```
plt.show()
```

<Figure size 800x500 with 0 Axes>



```
[23]: previous_default_analysis = (  
    data.groupby('previous_defaults')['default']  
        .agg(['count', 'sum', 'mean'])  
    )  
  
    previous_default_analysis['default_rate_%'] = (  
        previous_default_analysis['mean'] * 100  
    )  
  
    previous_default_analysis
```

```
[23]:
```

	count	sum	mean	default_rate_%
previous_defaults				
0	39743	3604	0.090683	9.068264
1	8979	995	0.110814	11.081412
2	1149	158	0.137511	13.751088
3	116	24	0.206897	20.689655

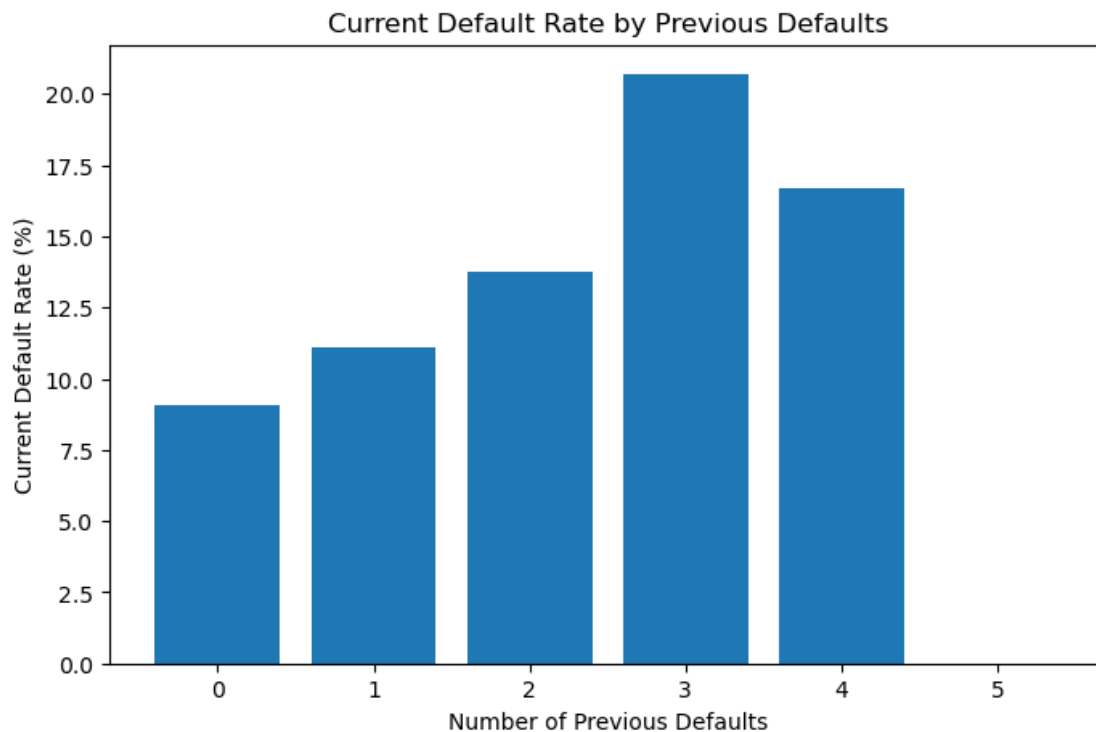
4	12	2	0.166667	16.666667
5	1	0	0.000000	0.000000

```
[24]: plt.figure(figsize=(8,5))

plt.bar(
    previous_default_analysis.index.astype(str),
    previous_default_analysis['default_rate_%']
)

plt.title('Current Default Rate by Previous Defaults')
plt.xlabel('Number of Previous Defaults')
plt.ylabel('Current Default Rate (%)')

plt.show()
```



```
[25]: # PREVIOUS LATE PAYMENTS VS CURRENT DEFAULT
late_payment_analysis = (
    data.groupby('previous_late_payments')['default']
        .agg(['count', 'sum', 'mean'])
)

late_payment_analysis['default_rate_%'] = (
```

```

    late_payment_analysis['mean'] * 100
)

late_payment_analysis

```

```

[25]:

```

	count	sum	mean	default_rate_%
previous_late_payments				
0	19986	1885	0.094316	9.431602
1	17595	1634	0.092867	9.286729
2	8545	851	0.099590	9.959040
3	2931	309	0.105425	10.542477
4	718	74	0.103064	10.306407
5	183	27	0.147541	14.754098
6	35	3	0.085714	8.571429
7	6	0	0.000000	0.000000
9	1	0	0.000000	0.000000

```

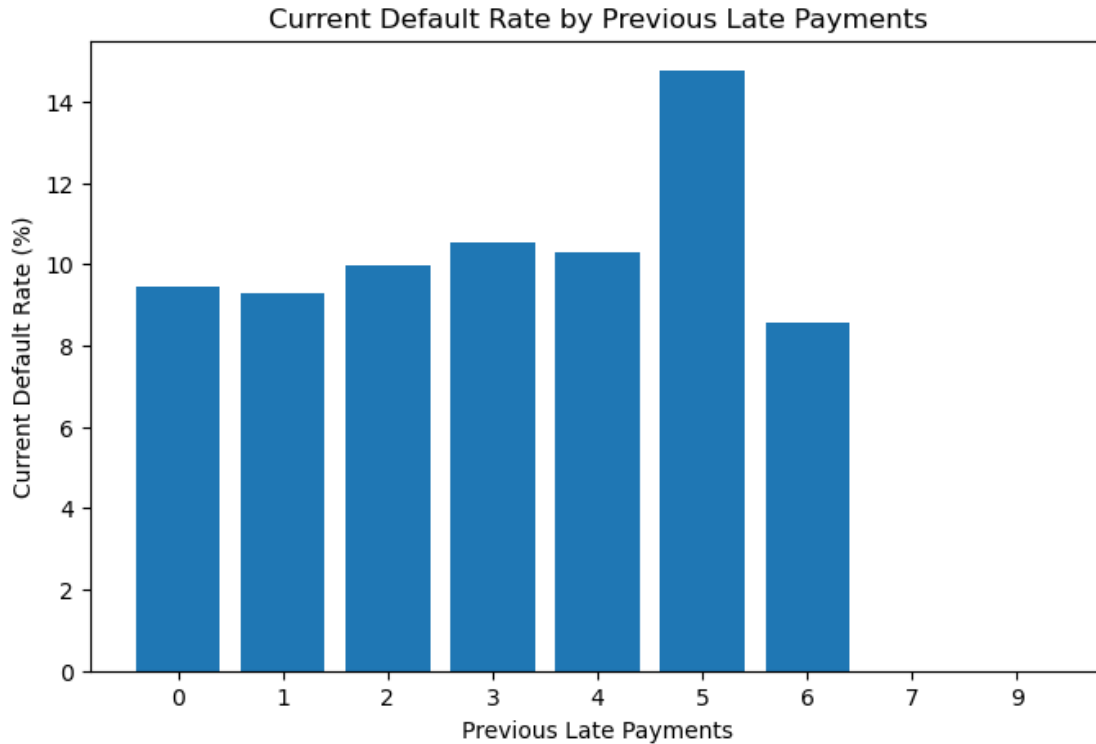
[26]: plt.figure(figsize=(8,5))

plt.bar(
    late_payment_analysis.index.astype(str),
    late_payment_analysis['default_rate_%']
)

plt.title('Current Default Rate by Previous Late Payments')
plt.xlabel('Previous Late Payments')
plt.ylabel('Current Default Rate (%)')

plt.show()

```



```
[27]: # SAVINGS BALANCE VS DEFAULT
data.groupby('default')['savings_balance_kes'].agg(
    ['count', 'mean', 'median', 'std', 'min', 'max']
)
```

```
[27]:
```

	count	mean	median	std	min	max
default						
0	45217	197987.913838	165800.0	140219.261901	1000	1459300
1	4783	133829.918461	93000.0	129240.907828	1000	893600

```
[28]: # OTHER DEBT VS DEFAULT
data.groupby('default')['other_monthly_debt_kes'].agg(
    ['count', 'mean', 'median', 'std', 'min', 'max']
)
```

```
[28]:
```

	count	mean	median	std	min	max
default						
0	45217	11897.722096	9900.0	8844.052260	300	99200
1	4783	12199.958185	10000.0	9026.593685	300	66300

```
[29]: # SEPARATE NUMERICAL AND CATEGORICAL VARIABLES
# Numerical columns
numerical_cols = data.select_dtypes(include=['int64', 'float64']).columns
```

```
numerical_cols
```

```
[29]: Index(['loan_id', 'borrower_id', 'age', 'monthly_income_kes',
          'membership_years', 'savings_balance_kes', 'loan_amount_kes',
          'loan_term_months', 'monthly_repayment_kes', 'other_monthly_debt_kes',
          'debt_to_income_ratio', 'loan_to_savings_ratio', 'previous_loans',
          'previous_defaults', 'previous_late_payments',
          'avg_monthly_inflows_kes', 'avg_monthly_outflows_kes',
          'mobile_money_active_months', 'utility_payment_consistency',
          'income_consistency_score', 'days_overdue',
          'late_payment_count_current', 'repayment_ratio', 'default',
          'generated_default_probability'],
         dtype='object')
```

```
[30]: # Categorical columns
categorical_cols = data.select_dtypes(include=['object']).columns

categorical_cols
```

```
[30]: Index(['application_date', 'branch', 'county', 'employment_type',
          'loan_product', 'loan_status', 'risk_category'],
         dtype='object')
```

```
[31]: # NUMERICAL UNIVARIATE ANALYSIS
data[numerical_cols].describe().T
```

```
[31]:
```

	count	mean	std	\
loan_id	50000.0	5.025000e+06	14433.901067	
borrower_id	50000.0	1.250005e+05	14433.901067	
age	50000.0	3.713746e+01	8.690509	
monthly_income_kes	50000.0	6.465726e+04	30738.530135	
membership_years	50000.0	5.009140e+00	3.164752	
savings_balance_kes	50000.0	1.918506e+05	140478.531476	
loan_amount_kes	50000.0	3.165825e+05	188116.668668	
loan_term_months	50000.0	1.848576e+01	14.069732	
monthly_repayment_kes	50000.0	3.698308e+04	43204.064126	
other_monthly_debt_kes	50000.0	1.192663e+04	8862.031096	
debt_to_income_ratio	50000.0	7.499959e-01	0.538381	
loan_to_savings_ratio	50000.0	3.039788e+00	10.106464	
previous_loans	50000.0	7.500560e+00	4.670815	
previous_defaults	50000.0	2.335600e-01	0.492112	
previous_late_payments	50000.0	9.505200e-01	1.006763	
avg_monthly_inflows_kes	50000.0	7.109106e+04	35248.188954	
avg_monthly_outflows_kes	50000.0	5.793037e+04	29783.851806	
mobile_money_active_months	50000.0	4.953524e+01	31.760548	
utility_payment_consistency	50000.0	8.133958e-01	0.128038	

income_consistency_score	50000.0	7.634962e-01	0.153210
days_overdue	50000.0	1.112082e+01	20.280348
late_payment_count_current	50000.0	8.055200e-01	1.193418
repayment_ratio	50000.0	8.816185e-01	0.139904
default	50000.0	9.566000e-02	0.294127
generated_default_probability	50000.0	9.610145e-02	0.150057

	min	25%	50% \
loan_id	5.000001e+06	5.012501e+06	5.025000e+06
borrower_id	1.000010e+05	1.125008e+05	1.250005e+05
age	2.100000e+01	3.100000e+01	3.700000e+01
monthly_income_kes	1.000000e+04	4.280000e+04	5.940000e+04
membership_years	0.000000e+00	3.000000e+00	4.000000e+00
savings_balance_kes	1.000000e+03	8.680000e+04	1.596000e+05
loan_amount_kes	1.300000e+04	1.797000e+05	2.739000e+05
loan_term_months	3.000000e+00	6.000000e+00	1.200000e+01
monthly_repayment_kes	3.000000e+02	1.060000e+04	2.180000e+04
other_monthly_debt_kes	3.000000e+02	5.200000e+03	9.900000e+03
debt_to_income_ratio	6.000000e-02	3.860000e-01	5.690000e-01
loan_to_savings_ratio	3.410000e-01	1.198000e+00	1.679000e+00
previous_loans	0.000000e+00	4.000000e+00	7.000000e+00
previous_defaults	0.000000e+00	0.000000e+00	0.000000e+00
previous_late_payments	0.000000e+00	0.000000e+00	1.000000e+00
avg_monthly_inflows_kes	8.900000e+03	4.610000e+04	6.480000e+04
avg_monthly_outflows_kes	6.000000e+03	3.680000e+04	5.240000e+04
mobile_money_active_months	3.000000e+00	2.600000e+01	4.300000e+01
utility_payment_consistency	2.500000e-01	7.250000e-01	8.200000e-01
income_consistency_score	2.500000e-01	6.590000e-01	7.690000e-01
days_overdue	0.000000e+00	2.000000e+00	5.000000e+00
late_payment_count_current	0.000000e+00	0.000000e+00	0.000000e+00
repayment_ratio	2.500000e-01	8.550000e-01	9.100000e-01
default	0.000000e+00	0.000000e+00	0.000000e+00
generated_default_probability	6.000000e-04	2.450000e-02	4.670000e-02

	75%	max
loan_id	5.037500e+06	5050000.00
borrower_id	1.375002e+05	150000.00
age	4.300000e+01	65.00
monthly_income_kes	8.060000e+04	365400.00
membership_years	7.000000e+00	20.00
savings_balance_kes	2.632250e+05	1459300.00
loan_amount_kes	4.081250e+05	1000000.00
loan_term_months	2.400000e+01	60.00
monthly_repayment_kes	4.580000e+04	406500.00
other_monthly_debt_kes	1.640000e+04	99200.00
debt_to_income_ratio	9.320000e-01	2.50
loan_to_savings_ratio	2.618000e+00	338.30

previous_loans	1.000000e+01	25.00
previous_defaults	0.000000e+00	5.00
previous_late_payments	1.000000e+00	9.00
avg_monthly_inflows_kes	8.890000e+04	451700.00
avg_monthly_outflows_kes	7.250000e+04	390300.00
mobile_money_active_months	6.600000e+01	180.00
utility_payment_consistency	9.150000e-01	1.00
income_consistency_score	8.800000e-01	1.00
days_overdue	9.000000e+00	251.00
late_payment_count_current	1.000000e+00	11.00
repayment_ratio	9.650000e-01	1.02
default	0.000000e+00	1.00
generated_default_probability	9.660000e-02	1.00

```
[32]: # CATEGORICAL UNIVARIATE ANALYSIS
for col in categorical_cols:
    print(f"\n{'='*50}")
    print(col)
    print(data[col].value_counts())
```

```
=====
application_date
application_date
2025-12-31    77
2026-01-07    75
2026-08-12    70
2024-10-31    69
2025-08-26    68
..
2025-09-28    34
2024-12-06    34
2024-07-05    34
2025-01-19    33
2025-03-28    33
Name: count, Length: 974, dtype: int64
```

```
=====
branch
branch
Nakuru        5115
Meru          5083
Nyeri         5051
Westlands     5050
Embu          4982
Nairobi CBD   4961
Eldoret       4961
Thika         4959
```

```
Kisumu          4922
Mombasa         4916
Name: count, dtype: int64
```

=====

```
county
county
Nairobi         12498
Kiambu          5060
Nakuru          4563
Other           4360
Mombasa         3519
Meru            3467
Machakos        3333
Uasin Gishu     3061
Kisumu          3033
Kakamega        2582
Embu            2516
Nyeri           2008
Name: count, dtype: int64
```

=====

```
employment_type
employment_type
Salaried        17091
Self-Employed   13479
Business Owner   9027
Farmer          5979
Casual          4424
Name: count, dtype: int64
```

=====

```
loan_product
loan_product
Development     11989
Business        9972
Emergency       7846
Agriculture     7113
School Fees     7048
Asset Finance   6032
Name: count, dtype: int64
```

=====

```
loan_status
loan_status
Late            28109
Current         17108
Defaulted       2656
```

```
Severely Delinquent      1376
Restructured             751
Name: count, dtype: int64
```

```
=====
risk_category
risk_category
Low          37961
Moderate     7914
High         2612
Very High    1513
Name: count, dtype: int64
```

```
[33]: correlation_matrix = data[numerical_cols].corr()
```

```
correlation_matrix
```

```
[33]:
```

	loan_id	borrower_id	age \
loan_id	1.000000	1.000000	-0.004792
borrower_id	1.000000	1.000000	-0.004792
age	-0.004792	-0.004792	1.000000
monthly_income_kes	-0.003064	-0.003064	-0.006274
membership_years	0.005236	0.005236	-0.008628
savings_balance_kes	0.000313	0.000313	-0.002004
loan_amount_kes	-0.003627	-0.003627	-0.008233
loan_term_months	0.001148	0.001148	0.011451
monthly_repayment_kes	-0.003821	-0.003821	-0.008385
other_monthly_debt_kes	-0.004442	-0.004442	-0.000776
debt_to_income_ratio	0.002013	0.002013	-0.009009
loan_to_savings_ratio	0.003416	0.003416	0.001453
previous_loans	0.005297	0.005297	-0.006206
previous_defaults	0.001090	0.001090	0.003132
previous_late_payments	0.010035	0.010035	-0.004391
avg_monthly_inflows_kes	-0.002888	-0.002888	-0.004809
avg_monthly_outflows_kes	-0.004000	-0.004000	-0.005411
mobile_money_active_months	0.005347	0.005347	-0.009856
utility_payment_consistency	0.003935	0.003935	-0.003523
income_consistency_score	0.000050	0.000050	-0.002564
days_overdue	0.004065	0.004065	-0.002188
late_payment_count_current	-0.001559	-0.001559	-0.003967
repayment_ratio	-0.004467	-0.004467	-0.002513
default	0.000564	0.000564	0.000458
generated_default_probability	-0.004527	-0.004527	0.000137

	monthly_income_kes	membership_years \
loan_id	-0.003064	0.005236
borrower_id	-0.003064	0.005236

age	-0.006274	-0.008628
monthly_income_kes	1.000000	0.003983
membership_years	0.003983	1.000000
savings_balance_kes	0.659443	0.002833
loan_amount_kes	0.767018	-0.001711
loan_term_months	0.003123	0.009862
monthly_repayment_kes	0.384906	-0.004817
other_monthly_debt_kes	0.639470	0.004825
debt_to_income_ratio	-0.013189	-0.003957
loan_to_savings_ratio	-0.063064	0.001859
previous_loans	0.004633	0.796446
previous_defaults	-0.000360	0.133150
previous_late_payments	0.003876	0.132555
avg_monthly_inflows_kes	0.955930	0.004033
avg_monthly_outflows_kes	0.924729	0.004274
mobile_money_active_months	0.004912	0.974602
utility_payment_consistency	-0.003492	0.000047
income_consistency_score	0.078326	-0.000406
days_overdue	-0.005315	-0.071853
late_payment_count_current	-0.009520	-0.055339
repayment_ratio	0.006964	0.077915
default	-0.007759	-0.089893
generated_default_probability	-0.022109	-0.173008

	savings_balance_kes	loan_amount_kes	\
loan_id	0.000313	-0.003627	
borrower_id	0.000313	-0.003627	
age	-0.002004	-0.008233	
monthly_income_kes	0.659443	0.767018	
membership_years	0.002833	-0.001711	
savings_balance_kes	1.000000	0.687144	
loan_amount_kes	0.687144	1.000000	
loan_term_months	0.000010	0.001090	
monthly_repayment_kes	0.343834	0.502782	
other_monthly_debt_kes	0.422052	0.491454	
debt_to_income_ratio	0.078984	0.190209	
loan_to_savings_ratio	-0.187786	-0.050350	
previous_loans	0.002768	0.000758	
previous_defaults	-0.001353	-0.001040	
previous_late_payments	0.002598	-0.000056	
avg_monthly_inflows_kes	0.629127	0.731993	
avg_monthly_outflows_kes	0.606379	0.707932	
mobile_money_active_months	0.004035	-0.001371	
utility_payment_consistency	-0.003063	-0.009185	
income_consistency_score	0.053352	0.061019	
days_overdue	-0.112087	0.042464	
late_payment_count_current	-0.090731	0.030671	

repayment_ratio	0.117736	-0.041036
default	-0.134331	0.049683
generated_default_probability	-0.265512	0.082953

	loan_term_months	monthly_repayment_kes \
loan_id	0.001148	-0.003821
borrower_id	0.001148	-0.003821
age	0.011451	-0.008385
monthly_income_kes	0.003123	0.384906
membership_years	0.009862	-0.004817
savings_balance_kes	0.000010	0.343834
loan_amount_kes	0.001090	0.502782
loan_term_months	1.000000	-0.536250
monthly_repayment_kes	-0.536250	1.000000
other_monthly_debt_kes	0.003365	0.242725
debt_to_income_ratio	-0.649644	0.813300
loan_to_savings_ratio	-0.004058	-0.026344
previous_loans	0.005625	-0.001118
previous_defaults	0.000469	0.001478
previous_late_payments	-0.000726	-0.000468
avg_monthly_inflows_kes	0.004082	0.363923
avg_monthly_outflows_kes	0.003018	0.353420
mobile_money_active_months	0.009195	-0.004322
utility_payment_consistency	-0.005994	0.005672
income_consistency_score	-0.007532	0.034395
days_overdue	-0.068622	0.114962
late_payment_count_current	-0.055054	0.088850
repayment_ratio	0.069212	-0.115393
default	-0.083739	0.137067
generated_default_probability	-0.171786	0.270418

	other_monthly_debt_kes ... \
loan_id	-0.004442 ...
borrower_id	-0.004442 ...
age	-0.000776 ...
monthly_income_kes	0.639470 ...
membership_years	0.004825 ...
savings_balance_kes	0.422052 ...
loan_amount_kes	0.491454 ...
loan_term_months	0.003365 ...
monthly_repayment_kes	0.242725 ...
other_monthly_debt_kes	1.000000 ...
debt_to_income_ratio	0.110678 ...
loan_to_savings_ratio	-0.041783 ...
previous_loans	0.006817 ...
previous_defaults	-0.001305 ...
previous_late_payments	0.008125 ...

avg_monthly_inflows_kes	0.610515	...
avg_monthly_outflows_kes	0.589822	...
mobile_money_active_months	0.004850	...
utility_payment_consistency	-0.001994	...
income_consistency_score	0.043630	...
days_overdue	0.008492	...
late_payment_count_current	0.001966	...
repayment_ratio	-0.009364	...
default	0.010031	...
generated_default_probability	0.008327	...

	avg_monthly_inflows_kes	\
loan_id	-0.002888	
borrower_id	-0.002888	
age	-0.004809	
monthly_income_kes	0.955930	
membership_years	0.004033	
savings_balance_kes	0.629127	
loan_amount_kes	0.731993	
loan_term_months	0.004082	
monthly_repayment_kes	0.363923	
other_monthly_debt_kes	0.610515	
debt_to_income_ratio	-0.015170	
loan_to_savings_ratio	-0.060383	
previous_loans	0.004522	
previous_defaults	0.000283	
previous_late_payments	0.001892	
avg_monthly_inflows_kes	1.000000	
avg_monthly_outflows_kes	0.967298	
mobile_money_active_months	0.005412	
utility_payment_consistency	-0.002769	
income_consistency_score	0.075696	
days_overdue	-0.006995	
late_payment_count_current	-0.010794	
repayment_ratio	0.007711	
default	-0.008931	
generated_default_probability	-0.021900	

	avg_monthly_outflows_kes	\
loan_id	-0.004000	
borrower_id	-0.004000	
age	-0.005411	
monthly_income_kes	0.924729	
membership_years	0.004274	
savings_balance_kes	0.606379	
loan_amount_kes	0.707932	
loan_term_months	0.003018	

monthly_repayment_kes	0.353420
other_monthly_debt_kes	0.589822
debt_to_income_ratio	-0.013499
loan_to_savings_ratio	-0.057232
previous_loans	0.005150
previous_defaults	0.000224
previous_late_payments	0.001711
avg_monthly_inflows_kes	0.967298
avg_monthly_outflows_kes	1.000000
mobile_money_active_months	0.005610
utility_payment_consistency	-0.002134
income_consistency_score	0.074192
days_overdue	-0.006203
late_payment_count_current	-0.010140
repayment_ratio	0.005984
default	-0.007259
generated_default_probability	-0.020111

	mobile_money_active_months \
loan_id	0.005347
borrower_id	0.005347
age	-0.009856
monthly_income_kes	0.004912
membership_years	0.974602
savings_balance_kes	0.004035
loan_amount_kes	-0.001371
loan_term_months	0.009195
monthly_repayment_kes	-0.004322
other_monthly_debt_kes	0.004850
debt_to_income_ratio	-0.004015
loan_to_savings_ratio	0.002036
previous_loans	0.777284
previous_defaults	0.129983
previous_late_payments	0.129437
avg_monthly_inflows_kes	0.005412
avg_monthly_outflows_kes	0.005610
mobile_money_active_months	1.000000
utility_payment_consistency	0.001316
income_consistency_score	0.000058
days_overdue	-0.070135
late_payment_count_current	-0.052699
repayment_ratio	0.074761
default	-0.087010
generated_default_probability	-0.169612

	utility_payment_consistency \
loan_id	0.003935

borrower_id	0.003935
age	-0.003523
monthly_income_kes	-0.003492
membership_years	0.000047
savings_balance_kes	-0.003063
loan_amount_kes	-0.009185
loan_term_months	-0.005994
monthly_repayment_kes	0.005672
other_monthly_debt_kes	-0.001994
debt_to_income_ratio	0.007330
loan_to_savings_ratio	-0.005042
previous_loans	0.001393
previous_defaults	-0.002055
previous_late_payments	0.006994
avg_monthly_inflows_kes	-0.002769
avg_monthly_outflows_kes	-0.002134
mobile_money_active_months	0.001316
utility_payment_consistency	1.000000
income_consistency_score	0.002093
days_overdue	-0.023962
late_payment_count_current	-0.017559
repayment_ratio	0.019965
default	-0.024988
generated_default_probability	-0.043728

	income_consistency_score	days_overdue \
loan_id	0.000050	0.004065
borrower_id	0.000050	0.004065
age	-0.002564	-0.002188
monthly_income_kes	0.078326	-0.005315
membership_years	-0.000406	-0.071853
savings_balance_kes	0.053352	-0.112087
loan_amount_kes	0.061019	0.042464
loan_term_months	-0.007532	-0.068622
monthly_repayment_kes	0.034395	0.114962
other_monthly_debt_kes	0.043630	0.008492
debt_to_income_ratio	0.001275	0.131568
loan_to_savings_ratio	-0.010807	0.244418
previous_loans	0.005173	-0.051943
previous_defaults	0.007640	0.033136
previous_late_payments	-0.002187	0.008906
avg_monthly_inflows_kes	0.075696	-0.006995
avg_monthly_outflows_kes	0.074192	-0.006203
mobile_money_active_months	0.000058	-0.070135
utility_payment_consistency	0.002093	-0.023962
income_consistency_score	1.000000	-0.028893
days_overdue	-0.028893	1.000000

late_payment_count_current
 repayment_ratio
 default
 generated_default_probability

-0.019822 0.557453
 0.029192 -0.712875
 -0.030104 0.838729
 -0.053513 0.429979

	late_payment_count_current	repayment_ratio \
loan_id	-0.001559	-0.004467
borrower_id	-0.001559	-0.004467
age	-0.003967	-0.002513
monthly_income_kes	-0.009520	0.006964
membership_years	-0.055339	0.077915
savings_balance_kes	-0.090731	0.117736
loan_amount_kes	0.030671	-0.041036
loan_term_months	-0.055054	0.069212
monthly_repayment_kes	0.088850	-0.115393
other_monthly_debt_kes	0.001966	-0.009364
debt_to_income_ratio	0.105320	-0.136639
loan_to_savings_ratio	0.189504	-0.238283
previous_loans	-0.040080	0.056586
previous_defaults	0.030927	-0.036102
previous_late_payments	0.011683	-0.007038
avg_monthly_inflows_kes	-0.010794	0.007711
avg_monthly_outflows_kes	-0.010140	0.005984
mobile_money_active_months	-0.052699	0.074761
utility_payment_consistency	-0.017559	0.019965
income_consistency_score	-0.019822	0.029192
days_overdue	0.557453	-0.712875
late_payment_count_current	1.000000	-0.558749
repayment_ratio	-0.558749	1.000000
default	0.660392	-0.854773
generated_default_probability	0.338412	-0.438135

	default	generated_default_probability
loan_id	0.000564	-0.004527
borrower_id	0.000564	-0.004527
age	0.000458	0.000137
monthly_income_kes	-0.007759	-0.022109
membership_years	-0.089893	-0.173008
savings_balance_kes	-0.134331	-0.265512
loan_amount_kes	0.049683	0.082953
loan_term_months	-0.083739	-0.171786
monthly_repayment_kes	0.137067	0.270418
other_monthly_debt_kes	0.010031	0.008327
debt_to_income_ratio	0.159163	0.317065
loan_to_savings_ratio	0.287249	0.561776
previous_loans	-0.066083	-0.126222
previous_defaults	0.037845	0.070269

previous_late_payments	0.011189	0.017308
avg_monthly_inflows_kes	-0.008931	-0.021900
avg_monthly_outflows_kes	-0.007259	-0.020111
mobile_money_active_months	-0.087010	-0.169612
utility_payment_consistency	-0.024988	-0.043728
income_consistency_score	-0.030104	-0.053513
days_overdue	0.838729	0.429979
late_payment_count_current	0.660392	0.338412
repayment_ratio	-0.854773	-0.438135
default	1.000000	0.513710
generated_default_probability	0.513710	1.000000

[25 rows x 25 columns]

```
[34]: data['default'].corr(data['loan_amount_kes'])
```

```
[34]: np.float64(0.04968311534903406)
```

```
[35]: data['default'].corr(data['previous_defaults'])
```

```
[35]: np.float64(0.03784454836033533)
```

```
[36]: data['default'].corr(data['previous_late_payments'])
```

```
[36]: np.float64(0.011189252641008871)
```

```
[37]: data['default'].corr(data['monthly_income_kes'])
```

```
[37]: np.float64(-0.007759138988570828)
```

```
[38]: data['default'].corr(data['other_monthly_debt_kes'])
```

```
[38]: np.float64(0.010031097543155244)
```

```
[39]: data['default'].corr(data['savings_balance_kes'])
```

```
[39]: np.float64(-0.13433099415476127)
```

```
[40]: #FEATURE ENGINEERING
#How large is the loan compared with income?
#Loan-to-Income Ratio

data['loan_to_income_ratio'] = (
    data['loan_amount_kes'] / data['monthly_income_kes']
)
```

```
[41]: # Debt-to-Income Ratio(DTI).
```

```
# Are borrowers who already have higher debt obligations relative to their
↳ income more likely to default?
data['debt_to_income_ratio'] = (
    data['other_monthly_debt_kes'] / data['monthly_income_kes']
)
```

```
[42]: # Total_Debt-to-Income Ratio(DTI).
# After taking this new loan, how much of the borrower's monthly income would
↳ be committed to debt payments?
data['total_debt_service_ratio'] = (
    (data['other_monthly_debt_kes'] + data['monthly_repayment_kes'])
    / data['monthly_income_kes']
)
```

```
[43]: # Monthly Cash Flow
# Do borrowers with weaker monthly cash flow have higher observed default rates?
data['monthly_cash_flow'] = (
    data['avg_monthly_inflows_kes']
    - data['avg_monthly_outflows_kes']
)
```

```
[44]: # Cash-Flow-to-Income Ratio
data['cash_flow_to_income_ratio'] = (
    data['monthly_cash_flow'] / data['monthly_income_kes']
)
```

```
[45]: # Repayment-to-Income Ratio
# Business question: Does a larger monthly repayment relative to income
↳ correspond with higher observed default rates?
data['repayment_to_income_ratio'] = (
    data['monthly_repayment_kes'] / data['monthly_income_kes']
)
```

```
[46]: # Savings-to-Income Ratio
# Do borrowers with stronger savings relative to income have different observed
↳ default rates?
data['savings_to_income_ratio'] = (
    data['savings_balance_kes'] / data['monthly_income_kes']
)
```

```
[47]: # Check the New Features
engineered_features = [
    'loan_to_income_ratio',
    'debt_to_income_ratio',
    'total_debt_service_ratio',
    'monthly_cash_flow',
    'cash_flow_to_income_ratio',
]
```

```

    'repayment_to_income_ratio',
    'savings_to_income_ratio'
]

data[engineered_features].describe().T

```

```

[47]:

```

	count	mean	std	min \
loan_to_income_ratio	50000.0	4.916455	1.739747	1.044077
debt_to_income_ratio	50000.0	0.184662	0.095575	0.018072
total_debt_service_ratio	50000.0	0.760318	0.577369	0.059937
monthly_cash_flow	50000.0	13160.684000	9925.769928	300.000000
cash_flow_to_income_ratio	50000.0	0.203907	0.108574	0.016598
repayment_to_income_ratio	50000.0	0.575656	0.569499	0.025180
savings_to_income_ratio	50000.0	2.960087	1.503106	0.013495

	25%	50%	75% \
loan_to_income_ratio	3.600000	4.787789	6.069642
debt_to_income_ratio	0.101852	0.184430	0.267559
total_debt_service_ratio	0.385538	0.569096	0.932259
monthly_cash_flow	5800.000000	10900.000000	18000.000000
cash_flow_to_income_ratio	0.111444	0.200508	0.289474
repayment_to_income_ratio	0.191980	0.374007	0.731524
savings_to_income_ratio	1.691205	2.968860	4.226686

	max
loan_to_income_ratio	12.492337
debt_to_income_ratio	0.352518
total_debt_service_ratio	4.552732
monthly_cash_flow	121200.000000
cash_flow_to_income_ratio	0.470948
repayment_to_income_ratio	4.512071
savings_to_income_ratio	7.721774

```

[48]: # Before interpreting the results, check whether your dataset contains any
      ↪ zero-income records.
      # If there are zero-income records, division can produce problematic values. If
      ↪ there aren't any, we're fine.
      (data['monthly_income_kes'] == 0).sum()

```

```

[48]: np.int64(0)

```

```

[49]: # COMPARE THESE FEATURES WITH DEFAULT
      data.groupby('default')[engineered_features].agg(
          ['mean', 'median']
      )

```

```
[49]:
```

	loan_to_income_ratio		debt_to_income_ratio		
	mean	median	mean	median	\
default					
0	4.873722	4.727903	0.183951	0.183333	
1	5.320432	5.306011	0.191389	0.194444	

	total_debt_service_ratio		monthly_cash_flow		
	mean	median	mean	median	\
default					
0	0.730187	0.554787	13192.754937	10900.0	
1	1.045164	0.778870	12857.495296	10400.0	

	cash_flow_to_income_ratio		repayment_to_income_ratio		
	mean	median	mean	median	\
default					
0	0.204285	0.200672		0.546237	
1	0.200335	0.198433		0.853775	

	savings_to_income_ratio		
	median	mean	median
default			
0	0.359107	3.057004	3.075697
1	0.576923	2.043861	1.586207

```
[50]: # BIVARIATE ANALYSIS ACROSS LEVELS, NOT JUST ON AVERAGES AND MEDIAN
```

```
data['repayment_ratio_group'] = pd.cut(
    data['repayment_to_income_ratio'],
    bins=[0, 0.30, 0.50, 0.70, 1.00, float('inf')],
    labels=[
        '0-30%',
        '30-50%',
        '50-70%',
        '70-100%',
        '100%+'
    ]
)

repayment_risk = (
    data.groupby('repayment_ratio_group', observed=True)['default']
    .agg(['count', 'sum', 'mean'])
)

repayment_risk['default_rate_%'] = repayment_risk['mean'] * 100

repayment_risk
```

```
[50]:
```

	count	sum	mean	default_rate_%
repayment_ratio_group				
0-30%	20785	1377	0.066250	6.624970
30-50%	9854	756	0.076720	7.672011
50-70%	6151	617	0.100309	10.030889
70-100%	4888	506	0.103519	10.351882
100%+	8322	1527	0.183490	18.348955

```
[51]: data['debt_service_group'] = pd.cut(
    data['total_debt_service_ratio'],
    bins=[0, 0.30, 0.50, 0.70, 1.00, float('inf')],
    labels=[
        '0-30%',
        '30-50%',
        '50-70%',
        '70-100%',
        '100%+'
    ]
)

debt_service_risk = (
    data.groupby('debt_service_group', observed=True)['default']
    .agg(['count', 'sum', 'mean'])
)

debt_service_risk['default_rate_%'] = (
    debt_service_risk['mean'] * 100
)

debt_service_risk
```

```
[51]:
```

	count	sum	mean	default_rate_%
debt_service_group				
0-30%	6835	430	0.062911	6.291149
30-50%	14046	951	0.067706	6.770611
50-70%	9960	746	0.074900	7.489960
70-100%	8003	811	0.101337	10.133700
100%+	11156	1845	0.165382	16.538186

```
[52]: data['savings_ratio_group'] = pd.cut(
    data['savings_to_income_ratio'],
    bins=[0, 1, 2, 3, float('inf')],
    labels=[
        '0-100%',
        '100-200%',
        '200-300%',
        '300%+'
    ]
)
```

```

    ]
)

savings_risk = (
    data.groupby('savings_ratio_group', observed=True)['default']
        .agg(['count', 'sum', 'mean'])
)

savings_risk['default_rate_%'] = (
    savings_risk['mean'] * 100
)

savings_risk

```

```

[52]:

```

	count	sum	mean	default_rate_%
savings_ratio_group				
0-100%	5932	1748	0.294673	29.467296
100-200%	9658	984	0.101884	10.188445
200-300%	9725	639	0.065707	6.570694
300%+	24685	1412	0.057201	5.720073

```

[53]: # MODEL FEATURES SELECTION

# remove the leakage columns from the predictive features
leakage_columns = [
    'days_overdue',
    'late_payment_count_current',
    'repayment_ratio',
    'loan_status',
    'generated_default_probability',
    'risk_category'
]

```

```

[54]: # remove the target
target = data['default']

```

```

[55]: # create our feature columns
model_feature_columns = [
    # Borrower profile
    'age',
    'employment_type',
    'county',
    'membership_years',

    # Financial capacity
    'monthly_income_kes',
    'savings_balance_kes',

```

```

'other_monthly_debt_kes',
'avg_monthly_inflows_kes',
'avg_monthly_outflows_kes',

# Loan characteristics
'loan_product',
'loan_amount_kes',
'loan_term_months',
'monthly_repayment_kes',

# Previous credit history
'previous_loans',
'previous_defaults',
'previous_late_payments',

# Financial/digital behavior
'mobile_money_active_months',
'utility_payment_consistency',
'income_consistency_score',

# Engineered features
'loan_to_income_ratio',
'debt_to_income_ratio',
'total_debt_service_ratio',
'monthly_cash_flow',
'cash_flow_to_income_ratio',
'repayment_to_income_ratio',
'savings_to_income_ratio'
]

x = data[model_feature_columns]
y = data['default']

```

```

[56]: # Check it:
x.shape

# it has 50000 rows and 26 columns instead of the original 32 columns
# we have removed 6 columns that would result in data leakage abd left
# with 26 columns/variables that can be used by the model for prediction

```

```
[56]: (50000, 26)
```

```
[57]: x.head()
```

```

[57]:   age employment_type      county membership_years  monthly_income_kes  \
0    21          Casual    Nairobi                2           28600
1    45        Salaried  Kakamega                5           50400

```

2	50	Business Owner	Nairobi	8	64200
3	27	Self-Employed	Kisumu	1	52900
4	53	Salaried	Uasin Gishu	2	117000

	savings_balance_kes	other_monthly_debt_kes	avg_monthly_inflows_kes	\
0	44900	800	27000	
1	91400	9000	50400	
2	174700	15700	84600	
3	223200	14700	48500	
4	309300	23800	157700	

	avg_monthly_outflows_kes	loan_product	...	mobile_money_active_months	\
0	25700	Agriculture	...	22	
1	35100	Development	...	53	
2	78700	Asset Finance	...	93	
3	35900	Business	...	11	
4	150500	Development	...	21	

	utility_payment_consistency	income_consistency_score	\
0	0.753	0.642	
1	0.754	0.940	
2	0.777	0.938	
3	0.747	0.735	
4	1.000	1.000	

	loan_to_income_ratio	debt_to_income_ratio	total_debt_service_ratio	\
0	2.388112	0.027972	0.444056	
1	7.208333	0.178571	0.414683	
2	3.931464	0.244548	0.434579	
3	6.395085	0.277883	0.884688	
4	5.725641	0.203419	0.391453	

	monthly_cash_flow	cash_flow_to_income_ratio	repayment_to_income_ratio	\
0	1300	0.045455	0.416084	
1	15300	0.303571	0.236111	
2	5900	0.091900	0.190031	
3	12600	0.238185	0.606805	
4	7200	0.061538	0.188034	

	savings_to_income_ratio
0	1.569930
1	1.813492
2	2.721184
3	4.219282
4	2.643590

[5 rows x 26 columns]

```
[58]: # check correlations among numerical features
```

```
numerical_features = x.select_dtypes(include=['int64', 'float64']).columns
model_feature_correlations = x[numerical_features].corr()
model_feature_correlations
```

```
[58]:
```

	age	membership_years	monthly_income_kes	\
age	1.000000	-0.008628	-0.006274	
membership_years	-0.008628	1.000000	0.003983	
monthly_income_kes	-0.006274	0.003983	1.000000	
savings_balance_kes	-0.002004	0.002833	0.659443	
other_monthly_debt_kes	-0.000776	0.004825	0.639470	
avg_monthly_inflows_kes	-0.004809	0.004033	0.955930	
avg_monthly_outflows_kes	-0.005411	0.004274	0.924729	
loan_amount_kes	-0.008233	-0.001711	0.767018	
loan_term_months	0.011451	0.009862	0.003123	
monthly_repayment_kes	-0.008385	-0.004817	0.384906	
previous_loans	-0.006206	0.796446	0.004633	
previous_defaults	0.003132	0.133150	-0.000360	
previous_late_payments	-0.004391	0.132555	0.003876	
mobile_money_active_months	-0.009856	0.974602	0.004912	
utility_payment_consistency	-0.003523	0.000047	-0.003492	
income_consistency_score	-0.002564	-0.000406	0.078326	
loan_to_income_ratio	-0.005055	-0.003883	-0.024347	
debt_to_income_ratio	0.003495	0.004299	-0.004464	
total_debt_service_ratio	-0.008502	-0.003760	-0.014107	
monthly_cash_flow	-0.000841	0.001498	0.619879	
cash_flow_to_income_ratio	0.002137	-0.001671	-0.007004	
repayment_to_income_ratio	-0.009206	-0.004534	-0.013553	
savings_to_income_ratio	0.001031	0.002057	0.009945	

	savings_balance_kes	other_monthly_debt_kes	\
age	-0.002004	-0.000776	
membership_years	0.002833	0.004825	
monthly_income_kes	0.659443	0.639470	
savings_balance_kes	1.000000	0.422052	
other_monthly_debt_kes	0.422052	1.000000	
avg_monthly_inflows_kes	0.629127	0.610515	
avg_monthly_outflows_kes	0.606379	0.589822	
loan_amount_kes	0.687144	0.491454	
loan_term_months	0.000010	0.003365	
monthly_repayment_kes	0.343834	0.242725	
previous_loans	0.002768	0.006817	
previous_defaults	-0.001353	-0.001305	
previous_late_payments	0.002598	0.008125	
mobile_money_active_months	0.004035	0.004850	
utility_payment_consistency	-0.003063	-0.001994	

income_consistency_score	0.053352	0.043630
loan_to_income_ratio	0.242692	-0.012798
debt_to_income_ratio	-0.004168	0.691995
total_debt_service_ratio	0.079691	0.104260
monthly_cash_flow	0.414609	0.398193
cash_flow_to_income_ratio	-0.001096	-0.002177
repayment_to_income_ratio	0.081491	-0.010433
savings_to_income_ratio	0.684095	0.005083

	avg_monthly_inflows_kes \
age	-0.004809
membership_years	0.004033
monthly_income_kes	0.955930
savings_balance_kes	0.629127
other_monthly_debt_kes	0.610515
avg_monthly_inflows_kes	1.000000
avg_monthly_outflows_kes	0.967298
loan_amount_kes	0.731993
loan_term_months	0.004082
monthly_repayment_kes	0.363923
previous_loans	0.004522
previous_defaults	0.000283
previous_late_payments	0.001892
mobile_money_active_months	0.005412
utility_payment_consistency	-0.002769
income_consistency_score	0.075696
loan_to_income_ratio	-0.024105
debt_to_income_ratio	-0.004116
total_debt_service_ratio	-0.015726
monthly_cash_flow	0.648646
cash_flow_to_income_ratio	0.057607
repayment_to_income_ratio	-0.015252
savings_to_income_ratio	0.007991

	avg_monthly_outflows_kes	loan_amount_kes \
age	-0.005411	-0.008233
membership_years	0.004274	-0.001711
monthly_income_kes	0.924729	0.767018
savings_balance_kes	0.606379	0.687144
other_monthly_debt_kes	0.589822	0.491454
avg_monthly_inflows_kes	0.967298	0.731993
avg_monthly_outflows_kes	1.000000	0.707932
loan_amount_kes	0.707932	1.000000
loan_term_months	0.003018	0.001090
monthly_repayment_kes	0.353420	0.502782
previous_loans	0.005150	0.000758
previous_defaults	0.000224	-0.001040

previous_late_payments	0.001711	-0.000056
mobile_money_active_months	0.005610	-0.001371
utility_payment_consistency	-0.002134	-0.009185
income_consistency_score	0.074192	0.061019
loan_to_income_ratio	-0.022964	0.565040
debt_to_income_ratio	-0.004713	-0.001189
total_debt_service_ratio	-0.014050	0.193738
monthly_cash_flow	0.434391	0.475176
cash_flow_to_income_ratio	-0.166631	-0.006742
repayment_to_income_ratio	-0.013453	0.196615
savings_to_income_ratio	0.006192	0.231385

	loan_term_months	monthly_repayment_kes	...	\
age	0.011451	-0.008385	...	
membership_years	0.009862	-0.004817	...	
monthly_income_kes	0.003123	0.384906	...	
savings_balance_kes	0.000010	0.343834	...	
other_monthly_debt_kes	0.003365	0.242725	...	
avg_monthly_inflows_kes	0.004082	0.363923	...	
avg_monthly_outflows_kes	0.003018	0.353420	...	
loan_amount_kes	0.001090	0.502782	...	
loan_term_months	1.000000	-0.536250	...	
monthly_repayment_kes	-0.536250	1.000000	...	
previous_loans	0.005625	-0.001118	...	
previous_defaults	0.000469	0.001478	...	
previous_late_payments	-0.000726	-0.000468	...	
mobile_money_active_months	0.009195	-0.004322	...	
utility_payment_consistency	-0.005994	0.005672	...	
income_consistency_score	-0.007532	0.034395	...	
loan_to_income_ratio	-0.001355	0.285785	...	
debt_to_income_ratio	0.003728	-0.003969	...	
total_debt_service_ratio	-0.625447	0.819938	...	
monthly_cash_flow	0.005440	0.231866	...	
cash_flow_to_income_ratio	0.002751	-0.009179	...	
repayment_to_income_ratio	-0.634716	0.831935	...	
savings_to_income_ratio	-0.001049	0.115280	...	

	mobile_money_active_months	\
age	-0.009856	
membership_years	0.974602	
monthly_income_kes	0.004912	
savings_balance_kes	0.004035	
other_monthly_debt_kes	0.004850	
avg_monthly_inflows_kes	0.005412	
avg_monthly_outflows_kes	0.005610	
loan_amount_kes	-0.001371	
loan_term_months	0.009195	

monthly_repayment_kes	-0.004322
previous_loans	0.777284
previous_defaults	0.129983
previous_late_payments	0.129437
mobile_money_active_months	1.000000
utility_payment_consistency	0.001316
income_consistency_score	0.000058
loan_to_income_ratio	-0.004480
debt_to_income_ratio	0.002310
total_debt_service_ratio	-0.003760
monthly_cash_flow	0.002383
cash_flow_to_income_ratio	-0.001262
repayment_to_income_ratio	-0.004200
savings_to_income_ratio	0.002275

	utility_payment_consistency \
age	-0.003523
membership_years	0.000047
monthly_income_kes	-0.003492
savings_balance_kes	-0.003063
other_monthly_debt_kes	-0.001994
avg_monthly_inflows_kes	-0.002769
avg_monthly_outflows_kes	-0.002134
loan_amount_kes	-0.009185
loan_term_months	-0.005994
monthly_repayment_kes	0.005672
previous_loans	0.001393
previous_defaults	-0.002055
previous_late_payments	0.006994
mobile_money_active_months	0.001316
utility_payment_consistency	1.000000
income_consistency_score	0.002093
loan_to_income_ratio	-0.006845
debt_to_income_ratio	0.000077
total_debt_service_ratio	0.006335
monthly_cash_flow	-0.003431
cash_flow_to_income_ratio	-0.005329
repayment_to_income_ratio	0.006410
savings_to_income_ratio	-0.000010

	income_consistency_score	loan_to_income_ratio \
age	-0.002564	-0.005055
membership_years	-0.000406	-0.003883
monthly_income_kes	0.078326	-0.024347
savings_balance_kes	0.053352	0.242692
other_monthly_debt_kes	0.043630	-0.012798
avg_monthly_inflows_kes	0.075696	-0.024105

avg_monthly_outflows_kes	0.074192	-0.022964
loan_amount_kes	0.061019	0.565040
loan_term_months	-0.007532	-0.001355
monthly_repayment_kes	0.034395	0.285785
previous_loans	0.005173	-0.002522
previous_defaults	0.007640	0.001160
previous_late_payments	-0.002187	-0.004767
mobile_money_active_months	0.000058	-0.004480
utility_payment_consistency	0.002093	-0.006845
income_consistency_score	1.000000	-0.004906
loan_to_income_ratio	-0.004906	1.000000
debt_to_income_ratio	-0.009069	0.005668
total_debt_service_ratio	0.000625	0.352355
monthly_cash_flow	0.046185	-0.016694
cash_flow_to_income_ratio	-0.004368	-0.002467
repayment_to_income_ratio	0.002156	0.356273
savings_to_income_ratio	0.002455	0.386863

	debt_to_income_ratio	total_debt_service_ratio \
age	0.003495	-0.008502
membership_years	0.004299	-0.003760
monthly_income_kes	-0.004464	-0.014107
savings_balance_kes	-0.004168	0.079691
other_monthly_debt_kes	0.691995	0.104260
avg_monthly_inflows_kes	-0.004116	-0.015726
avg_monthly_outflows_kes	-0.004713	-0.014050
loan_amount_kes	-0.001189	0.193738
loan_term_months	0.003728	-0.625447
monthly_repayment_kes	-0.003969	0.819938
previous_loans	0.005592	-0.001062
previous_defaults	0.001426	-0.001242
previous_late_payments	0.009496	0.001048
mobile_money_active_months	0.002310	-0.003760
utility_payment_consistency	0.000077	0.006335
income_consistency_score	-0.009069	0.000625
loan_to_income_ratio	0.005668	0.352355
debt_to_income_ratio	1.000000	0.164549
total_debt_service_ratio	0.164549	1.000000
monthly_cash_flow	-0.000477	-0.013686
cash_flow_to_income_ratio	0.001864	-0.007277
repayment_to_income_ratio	-0.001001	0.986204
savings_to_income_ratio	-0.002867	0.134090

	monthly_cash_flow	cash_flow_to_income_ratio \
age	-0.000841	0.002137
membership_years	0.001498	-0.001671
monthly_income_kes	0.619879	-0.007004

savings_balance_kes	0.414609	-0.001096
other_monthly_debt_kes	0.398193	-0.002177
avg_monthly_inflows_kes	0.648646	0.057607
avg_monthly_outflows_kes	0.434391	-0.166631
loan_amount_kes	0.475176	-0.006742
loan_term_months	0.005440	0.002751
monthly_repayment_kes	0.231866	-0.009179
previous_loans	0.000605	-0.005298
previous_defaults	0.000336	0.001630
previous_late_payments	0.001587	-0.001797
mobile_money_active_months	0.002383	-0.001262
utility_payment_consistency	-0.003431	-0.005329
income_consistency_score	0.046185	-0.004368
loan_to_income_ratio	-0.016694	-0.002467
debt_to_income_ratio	-0.000477	0.001864
total_debt_service_ratio	-0.013686	-0.007277
monthly_cash_flow	1.000000	0.704577
cash_flow_to_income_ratio	0.704577	1.000000
repayment_to_income_ratio	-0.013795	-0.007690
savings_to_income_ratio	0.009797	0.003293

	repayment_to_income_ratio \
age	-0.009206
membership_years	-0.004534
monthly_income_kes	-0.013553
savings_balance_kes	0.081491
other_monthly_debt_kes	-0.010433
avg_monthly_inflows_kes	-0.015252
avg_monthly_outflows_kes	-0.013453
loan_amount_kes	0.196615
loan_term_months	-0.634716
monthly_repayment_kes	0.831935
previous_loans	-0.002015
previous_defaults	-0.001498
previous_late_payments	-0.000531
mobile_money_active_months	-0.004200
utility_payment_consistency	0.006410
income_consistency_score	0.002156
loan_to_income_ratio	0.356273
debt_to_income_ratio	-0.001001
total_debt_service_ratio	0.986204
monthly_cash_flow	-0.013795
cash_flow_to_income_ratio	-0.007690
repayment_to_income_ratio	1.000000
savings_to_income_ratio	0.136424

savings_to_income_ratio

age	0.001031
membership_years	0.002057
monthly_income_kes	0.009945
savings_balance_kes	0.684095
other_monthly_debt_kes	0.005083
avg_monthly_inflows_kes	0.007991
avg_monthly_outflows_kes	0.006192
loan_amount_kes	0.231385
loan_term_months	-0.001049
monthly_repayment_kes	0.115280
previous_loans	-0.000509
previous_defaults	0.001929
previous_late_payments	0.001963
mobile_money_active_months	0.002275
utility_payment_consistency	-0.000010
income_consistency_score	0.002455
loan_to_income_ratio	0.386863
debt_to_income_ratio	-0.002867
total_debt_service_ratio	0.134090
monthly_cash_flow	0.009797
cash_flow_to_income_ratio	0.003293
repayment_to_income_ratio	0.136424
savings_to_income_ratio	1.000000

[23 rows x 23 columns]

```
[59]: import matplotlib.pyplot as plt

plt.figure(figsize=(14, 10))

plt.imshow(
    model_feature_correlations,
    cmap='coolwarm',
    aspect='auto'
)

plt.colorbar()

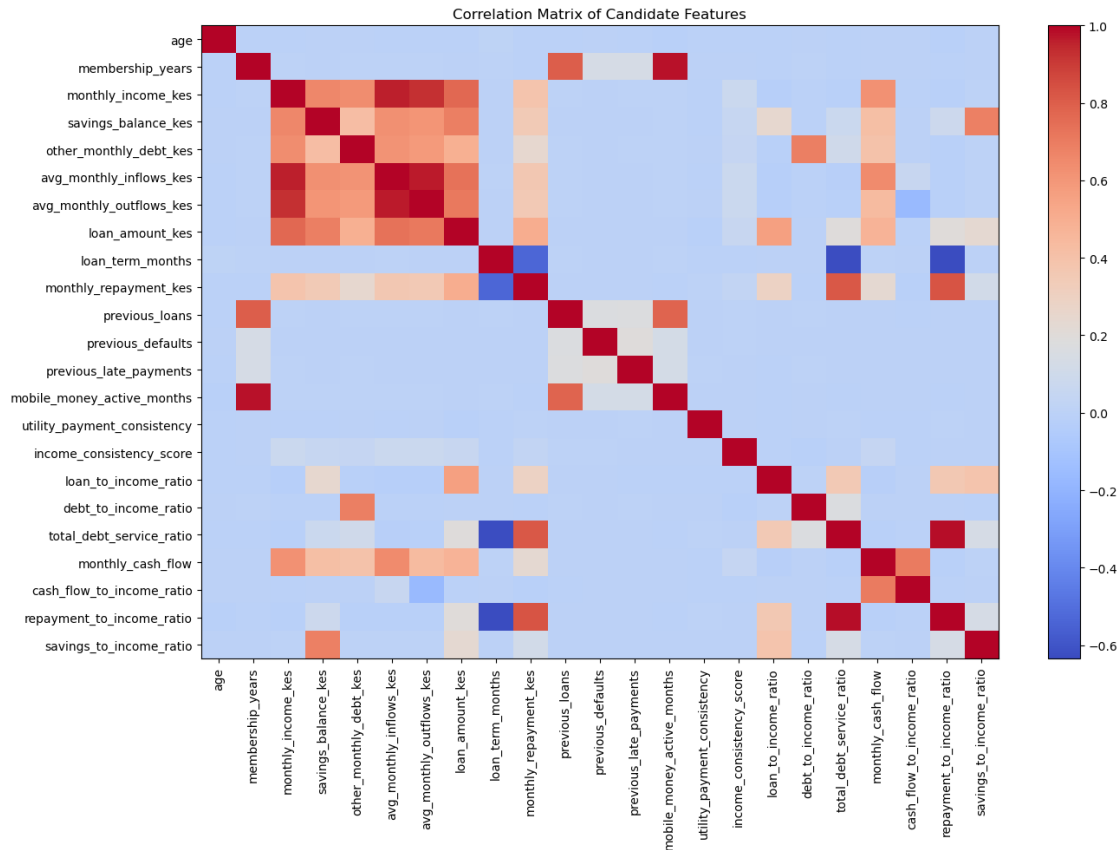
plt.xticks(
    range(len(model_feature_correlations.columns)),
    model_feature_correlations.columns,
    rotation=90
)

plt.yticks(
    range(len(model_feature_correlations.columns)),
    model_feature_correlations.columns
)
```

```
)

plt.title('Correlation Matrix of Candidate Features')

plt.tight_layout()
plt.show()
```



```
[60]: # Identify highly correlated feature pairs
highly_correlated_features_pairs = (
    model_feature_correlations.abs()
    .where(model_feature_correlations.abs() > 0.80)
)

highly_correlated_features_pairs
```

```
[60]:
```

	age	membership_years	monthly_income_kes	\
age	1.0	NaN	NaN	
membership_years	NaN	1.000000	NaN	
monthly_income_kes	NaN	NaN	1.000000	
savings_balance_kes	NaN	NaN	NaN	

other_monthly_debt_kes	NaN	NaN	NaN
avg_monthly_inflows_kes	NaN	NaN	0.955930
avg_monthly_outflows_kes	NaN	NaN	0.924729
loan_amount_kes	NaN	NaN	NaN
loan_term_months	NaN	NaN	NaN
monthly_repayment_kes	NaN	NaN	NaN
previous_loans	NaN	NaN	NaN
previous_defaults	NaN	NaN	NaN
previous_late_payments	NaN	NaN	NaN
mobile_money_active_months	NaN	0.974602	NaN
utility_payment_consistency	NaN	NaN	NaN
income_consistency_score	NaN	NaN	NaN
loan_to_income_ratio	NaN	NaN	NaN
debt_to_income_ratio	NaN	NaN	NaN
total_debt_service_ratio	NaN	NaN	NaN
monthly_cash_flow	NaN	NaN	NaN
cash_flow_to_income_ratio	NaN	NaN	NaN
repayment_to_income_ratio	NaN	NaN	NaN
savings_to_income_ratio	NaN	NaN	NaN

	savings_balance_kes	other_monthly_debt_kes	\
age	NaN	NaN	
membership_years	NaN	NaN	
monthly_income_kes	NaN	NaN	
savings_balance_kes	1.0	NaN	
other_monthly_debt_kes	NaN	1.0	
avg_monthly_inflows_kes	NaN	NaN	
avg_monthly_outflows_kes	NaN	NaN	
loan_amount_kes	NaN	NaN	
loan_term_months	NaN	NaN	
monthly_repayment_kes	NaN	NaN	
previous_loans	NaN	NaN	
previous_defaults	NaN	NaN	
previous_late_payments	NaN	NaN	
mobile_money_active_months	NaN	NaN	
utility_payment_consistency	NaN	NaN	
income_consistency_score	NaN	NaN	
loan_to_income_ratio	NaN	NaN	
debt_to_income_ratio	NaN	NaN	
total_debt_service_ratio	NaN	NaN	
monthly_cash_flow	NaN	NaN	
cash_flow_to_income_ratio	NaN	NaN	
repayment_to_income_ratio	NaN	NaN	
savings_to_income_ratio	NaN	NaN	

	avg_monthly_inflows_kes	\
age	NaN	

membership_years	NaN
monthly_income_kes	0.955930
savings_balance_kes	NaN
other_monthly_debt_kes	NaN
avg_monthly_inflows_kes	1.000000
avg_monthly_outflows_kes	0.967298
loan_amount_kes	NaN
loan_term_months	NaN
monthly_repayment_kes	NaN
previous_loans	NaN
previous_defaults	NaN
previous_late_payments	NaN
mobile_money_active_months	NaN
utility_payment_consistency	NaN
income_consistency_score	NaN
loan_to_income_ratio	NaN
debt_to_income_ratio	NaN
total_debt_service_ratio	NaN
monthly_cash_flow	NaN
cash_flow_to_income_ratio	NaN
repayment_to_income_ratio	NaN
savings_to_income_ratio	NaN

	avg_monthly_outflows_kes	loan_amount_kes	\
age	NaN	NaN	
membership_years	NaN	NaN	
monthly_income_kes	0.924729	NaN	
savings_balance_kes	NaN	NaN	
other_monthly_debt_kes	NaN	NaN	
avg_monthly_inflows_kes	0.967298	NaN	
avg_monthly_outflows_kes	1.000000	NaN	
loan_amount_kes	NaN	1.0	
loan_term_months	NaN	NaN	
monthly_repayment_kes	NaN	NaN	
previous_loans	NaN	NaN	
previous_defaults	NaN	NaN	
previous_late_payments	NaN	NaN	
mobile_money_active_months	NaN	NaN	
utility_payment_consistency	NaN	NaN	
income_consistency_score	NaN	NaN	
loan_to_income_ratio	NaN	NaN	
debt_to_income_ratio	NaN	NaN	
total_debt_service_ratio	NaN	NaN	
monthly_cash_flow	NaN	NaN	
cash_flow_to_income_ratio	NaN	NaN	
repayment_to_income_ratio	NaN	NaN	
savings_to_income_ratio	NaN	NaN	

	loan_term_months	monthly_repayment_kes	...	\
age	NaN	NaN	...	
membership_years	NaN	NaN	...	
monthly_income_kes	NaN	NaN	...	
savings_balance_kes	NaN	NaN	...	
other_monthly_debt_kes	NaN	NaN	...	
avg_monthly_inflows_kes	NaN	NaN	...	
avg_monthly_outflows_kes	NaN	NaN	...	
loan_amount_kes	NaN	NaN	...	
loan_term_months	1.0	NaN	...	
monthly_repayment_kes	NaN	1.000000	...	
previous_loans	NaN	NaN	...	
previous_defaults	NaN	NaN	...	
previous_late_payments	NaN	NaN	...	
mobile_money_active_months	NaN	NaN	...	
utility_payment_consistency	NaN	NaN	...	
income_consistency_score	NaN	NaN	...	
loan_to_income_ratio	NaN	NaN	...	
debt_to_income_ratio	NaN	NaN	...	
total_debt_service_ratio	NaN	0.819938	...	
monthly_cash_flow	NaN	NaN	...	
cash_flow_to_income_ratio	NaN	NaN	...	
repayment_to_income_ratio	NaN	0.831935	...	
savings_to_income_ratio	NaN	NaN	...	

	mobile_money_active_months	\
age	NaN	
membership_years	0.974602	
monthly_income_kes	NaN	
savings_balance_kes	NaN	
other_monthly_debt_kes	NaN	
avg_monthly_inflows_kes	NaN	
avg_monthly_outflows_kes	NaN	
loan_amount_kes	NaN	
loan_term_months	NaN	
monthly_repayment_kes	NaN	
previous_loans	NaN	
previous_defaults	NaN	
previous_late_payments	NaN	
mobile_money_active_months	1.000000	
utility_payment_consistency	NaN	
income_consistency_score	NaN	
loan_to_income_ratio	NaN	
debt_to_income_ratio	NaN	
total_debt_service_ratio	NaN	
monthly_cash_flow	NaN	

cash_flow_to_income_ratio	NaN
repayment_to_income_ratio	NaN
savings_to_income_ratio	NaN

	utility_payment_consistency	\
age	NaN	
membership_years	NaN	
monthly_income_kes	NaN	
savings_balance_kes	NaN	
other_monthly_debt_kes	NaN	
avg_monthly_inflows_kes	NaN	
avg_monthly_outflows_kes	NaN	
loan_amount_kes	NaN	
loan_term_months	NaN	
monthly_repayment_kes	NaN	
previous_loans	NaN	
previous_defaults	NaN	
previous_late_payments	NaN	
mobile_money_active_months	NaN	
utility_payment_consistency	1.0	
income_consistency_score	NaN	
loan_to_income_ratio	NaN	
debt_to_income_ratio	NaN	
total_debt_service_ratio	NaN	
monthly_cash_flow	NaN	
cash_flow_to_income_ratio	NaN	
repayment_to_income_ratio	NaN	
savings_to_income_ratio	NaN	

	income_consistency_score	loan_to_income_ratio	\
age	NaN	NaN	
membership_years	NaN	NaN	
monthly_income_kes	NaN	NaN	
savings_balance_kes	NaN	NaN	
other_monthly_debt_kes	NaN	NaN	
avg_monthly_inflows_kes	NaN	NaN	
avg_monthly_outflows_kes	NaN	NaN	
loan_amount_kes	NaN	NaN	
loan_term_months	NaN	NaN	
monthly_repayment_kes	NaN	NaN	
previous_loans	NaN	NaN	
previous_defaults	NaN	NaN	
previous_late_payments	NaN	NaN	
mobile_money_active_months	NaN	NaN	
utility_payment_consistency	NaN	NaN	
income_consistency_score	1.0	NaN	
loan_to_income_ratio	NaN	1.0	

debt_to_income_ratio	NaN	NaN
total_debt_service_ratio	NaN	NaN
monthly_cash_flow	NaN	NaN
cash_flow_to_income_ratio	NaN	NaN
repayment_to_income_ratio	NaN	NaN
savings_to_income_ratio	NaN	NaN

	debt_to_income_ratio	total_debt_service_ratio	\
age	NaN	NaN	
membership_years	NaN	NaN	
monthly_income_kes	NaN	NaN	
savings_balance_kes	NaN	NaN	
other_monthly_debt_kes	NaN	NaN	
avg_monthly_inflows_kes	NaN	NaN	
avg_monthly_outflows_kes	NaN	NaN	
loan_amount_kes	NaN	NaN	
loan_term_months	NaN	NaN	
monthly_repayment_kes	NaN	0.819938	
previous_loans	NaN	NaN	
previous_defaults	NaN	NaN	
previous_late_payments	NaN	NaN	
mobile_money_active_months	NaN	NaN	
utility_payment_consistency	NaN	NaN	
income_consistency_score	NaN	NaN	
loan_to_income_ratio	NaN	NaN	
debt_to_income_ratio	1.0	NaN	
total_debt_service_ratio	NaN	1.000000	
monthly_cash_flow	NaN	NaN	
cash_flow_to_income_ratio	NaN	NaN	
repayment_to_income_ratio	NaN	0.986204	
savings_to_income_ratio	NaN	NaN	

	monthly_cash_flow	cash_flow_to_income_ratio	\
age	NaN	NaN	
membership_years	NaN	NaN	
monthly_income_kes	NaN	NaN	
savings_balance_kes	NaN	NaN	
other_monthly_debt_kes	NaN	NaN	
avg_monthly_inflows_kes	NaN	NaN	
avg_monthly_outflows_kes	NaN	NaN	
loan_amount_kes	NaN	NaN	
loan_term_months	NaN	NaN	
monthly_repayment_kes	NaN	NaN	
previous_loans	NaN	NaN	
previous_defaults	NaN	NaN	
previous_late_payments	NaN	NaN	
mobile_money_active_months	NaN	NaN	

utility_payment_consistency	NaN	NaN
income_consistency_score	NaN	NaN
loan_to_income_ratio	NaN	NaN
debt_to_income_ratio	NaN	NaN
total_debt_service_ratio	NaN	NaN
monthly_cash_flow	1.0	NaN
cash_flow_to_income_ratio	NaN	1.0
repayment_to_income_ratio	NaN	NaN
savings_to_income_ratio	NaN	NaN

	repayment_to_income_ratio \	
age	NaN	
membership_years	NaN	
monthly_income_kes	NaN	
savings_balance_kes	NaN	
other_monthly_debt_kes	NaN	
avg_monthly_inflows_kes	NaN	
avg_monthly_outflows_kes	NaN	
loan_amount_kes	NaN	
loan_term_months	NaN	
monthly_repayment_kes	0.831935	
previous_loans	NaN	
previous_defaults	NaN	
previous_late_payments	NaN	
mobile_money_active_months	NaN	
utility_payment_consistency	NaN	
income_consistency_score	NaN	
loan_to_income_ratio	NaN	
debt_to_income_ratio	NaN	
total_debt_service_ratio	0.986204	
monthly_cash_flow	NaN	
cash_flow_to_income_ratio	NaN	
repayment_to_income_ratio	1.000000	
savings_to_income_ratio	NaN	

	savings_to_income_ratio	
age	NaN	
membership_years	NaN	
monthly_income_kes	NaN	
savings_balance_kes	NaN	
other_monthly_debt_kes	NaN	
avg_monthly_inflows_kes	NaN	
avg_monthly_outflows_kes	NaN	
loan_amount_kes	NaN	
loan_term_months	NaN	
monthly_repayment_kes	NaN	
previous_loans	NaN	

previous_defaults	NaN
previous_late_payments	NaN
mobile_money_active_months	NaN
utility_payment_consistency	NaN
income_consistency_score	NaN
loan_to_income_ratio	NaN
debt_to_income_ratio	NaN
total_debt_service_ratio	NaN
monthly_cash_flow	NaN
cash_flow_to_income_ratio	NaN
repayment_to_income_ratio	NaN
savings_to_income_ratio	1.0

[23 rows x 23 columns]

```
[61]: #Install/import VIF
from statsmodels.stats.outliers_influence import variance_inflation_factor
```

```
[62]: # Select the numerical model features
numerical_features = x.select_dtypes(
    include=['int64', 'float64']
).columns

x_numeric = x[numerical_features]
```

```
[63]: # calculate vif
vif = pd.DataFrame()

vif['feature'] = x_numeric.columns

vif['VIF'] = [
    variance_inflation_factor(
        x_numeric.values,
        i
    )
    for i in range(x_numeric.shape[1])
]

vif = vif.sort_values('VIF', ascending=False)

vif
```

```
C:\Users\User\anaconda3\Lib\site-
packages\statsmodels\stats\outliers_influence.py:197: RuntimeWarning: divide by
zero encountered in scalar divide
    vif = 1. / (1. - r_squared_i)
```

[63]:	feature	VIF
21	repayment_to_income_ratio	inf
19	monthly_cash_flow	inf
18	total_debt_service_ratio	inf
5	avg_monthly_inflows_kes	inf
6	avg_monthly_outflows_kes	inf
17	debt_to_income_ratio	inf
2	monthly_income_kes	111.648838
1	membership_years	75.693431
13	mobile_money_active_months	68.472572
7	loan_amount_kes	62.183513
16	loan_to_income_ratio	60.877871
14	utility_payment_consistency	32.262164
3	savings_balance_kes	30.634988
22	savings_to_income_ratio	29.663128
4	other_monthly_debt_kes	23.612846
15	income_consistency_score	22.577363
20	cash_flow_to_income_ratio	22.029024
0	age	17.285316
9	monthly_repayment_kes	13.076045
10	previous_loans	9.963031
8	loan_term_months	4.977220
12	previous_late_payments	2.003823
11	previous_defaults	1.298466

```
[65]: model_features = [
    # Borrower profile
    'age',
    'employment_type',
    'county',
    'membership_years',

    # Financial capacity
    'monthly_income_kes',
    'monthly_cash_flow',
    'savings_to_income_ratio',

    # Loan characteristics
    'loan_to_income_ratio',
    'loan_term_months',
    'monthly_repayment_kes',

    # Previous credit history
    'previous_loans',
    'previous_defaults',
    'previous_late_payments',
```

```

# Financial/digital behavior
'utility_payment_consistency',
'income_consistency_score',

# Engineered overall debt burden
'total_debt_service_ratio'
]

x_model = data[model_features]
y = data['default']

x_model.head()

# The training and testing data set(x) that the model will use are the features
↳ we have selected
# after first doing basic bivariate analysis between selected variables among
↳ the 31 variables/columns and the target variable and looking for correlations
# then engineering more features from the the 31 variables that will answer key
↳ questions during loan application time by borrowers
# then doing bivariate analysis between the engineered features and the target
↳ variable and looking for their correlations.
# correlations show the relationship between the features and the target,
↳ whether its positive or negative or no relationship at all
# we then do feature selection for the features the model will use. The model
↳ can not use all the 31 columns to do prediction during the application time
# it needs variables that will help predict default during the application time
# we tested for multicollinearity; identify variables that provide the same
↳ information(overlapping) to the model that will make it hard for the model to
↳ choose which variable to work with
# we also did vif analysis to further identify the best columns to add to the
↳ model
# many times, we eliminated variables that were used to form a final ratio/
↳ feature because they will just overlap with final feature itself

```

```

[65]:  age employment_type      county membership_years  monthly_income_kes  \
0    21          Casual      Nairobi              2           28600
1    45        Salaried    Kakamega              5           50400
2    50  Business Owner      Nairobi              8           64200
3    27  Self-Employed      Kisumu              1           52900
4    53        Salaried  Uasin Gishu              2          117000

      monthly_cash_flow  savings_to_income_ratio  loan_to_income_ratio  \
0              1300          1.569930          2.388112
1             15300          1.813492          7.208333
2              5900          2.721184          3.931464
3             12600          4.219282          6.395085

```

4	7200	2.643590	5.725641
---	------	----------	----------

	loan_term_months	monthly_repayment_kes	previous_loans	previous_defaults \
0	6	11900	4	0
1	36	11900	9	0
2	24	12200	16	0
3	12	32100	6	0
4	36	22000	4	0

	previous_late_payments	utility_payment_consistency \
0	1	0.753
1	1	0.754
2	0	0.777
3	2	0.747
4	2	1.000

	income_consistency_score	total_debt_service_ratio
0	0.642	0.444056
1	0.940	0.414683
2	0.938	0.434579
3	0.735	0.884688
4	1.000	0.391453

```
[67]: # check the shape of the model training features(x_model)
x_model.shape

# it has 50000 rows/borrowers and 16 prediction columns/variables
```

```
[67]: (50000, 16)
```

```
[68]: # Separate numerical and categorical variables
model_numerical_features = x_model.select_dtypes(include=['int64', 'float64']).
↳ columns
```

```
[69]: model_categorical_features = x_model.select_dtypes(include=['object']).columns
```

```
[72]: print('Model Numerical Features:')
print(list(model_numerical_features))

print('\nModel Categorical Features:')
print(list(model_categorical_features))
```

```
Model Numerical Features:
['age', 'membership_years', 'monthly_income_kes', 'monthly_cash_flow',
'savings_to_income_ratio', 'loan_to_income_ratio', 'loan_term_months',
'monthly_repayment_kes', 'previous_loans', 'previous_defaults',
'previous_late_payments', 'utility_payment_consistency',
'income_consistency_score', 'total_debt_service_ratio']
```

Model Categorical Features:
['employment_type', 'county']

```
[73]: # Split into training and testing data
from sklearn.model_selection import train_test_split

# We'll use 80% training and 20% testing.

x_train, x_test, y_train, y_test = train_test_split(
    x_model,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
```

```
[74]: # Because only about 9.57% of your borrowers defaulted, we want the training
      ↪ and testing datasets to maintain approximately the same default proportion.
print('Training default rate:')
print(y_train.mean())

print('\nTesting default rate:')
print(y_test.mean())
```

Training default rate:
0.09565

Testing default rate:
0.0957

```
[79]: # ENCODE THE MODEL CATEGORICAL VARIABLES AND SCALE THE MODEL NUMERICAL COLUMNS
      # Rather than manually encoding and scaling everything, we'll use
      ↪ Scikit-learn's ColumnTransformer and Pipeline. This is the cleaner
      ↪ professional approach because it prevents preprocessing information from
      ↪ accidentally leaking from the test set into the training process.
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
```

```
[80]: preprocessor = ColumnTransformer(
    transformers=[
        (
            'num',
            StandardScaler(),
            model_numerical_features
        ),
    ],
```

```

        (
            'cat',
            OneHotEncoder(
                handle_unknown='ignore',
                drop='first'
            ),
            model_categorical_features
        )
    ]
)

# The preprocessing pipeline: Scales numerical variables, Converts categorical
↪ variables into 0/1 columns
# and Applies exactly the same transformations to future/test data

```

```

[81]: # Build the Logistic Regression model
      from sklearn.linear_model import LogisticRegression

```

```

[82]: # Because only about 9.57% of borrowers default, we'll initially use:
      class_weight='balanced'
      # This tells Logistic Regression to give more importance to the minority class
      ↪ (default = 1).

```

```

[83]: # Create the pipeline:
      logistic_model = Pipeline(
          steps=[
              ('preprocessor', preprocessor),
              (
                  'classifier',
                  LogisticRegression(
                      class_weight='balanced',
                      max_iter=1000,
                      random_state=42
                  )
              )
          ]
      )

```

```

[85]: # Then train it:
      logistic_model.fit(x_train, y_train)

```

```

[85]: Pipeline(steps=[('preprocessor',
                        ColumnTransformer(transformers=[('num', StandardScaler(),
                                                         Index(['age',
                                                         'membership_years', 'monthly_income_kes', 'monthly_cash_flow',
                                                         'savings_to_income_ratio', 'loan_to_income_ratio', 'loan_term_months',
                                                         'monthly_repayment_kes', 'previous_loans', 'previous_defaults',

```

```

        'previous_late_payments', 'utility_payment_consistency',
        'income_consistency_score', 'total_debt_service_ratio'],
        dtype='object')),
        ('cat',
         OneHotEncoder(drop='first',
                        handle_unknown='ignore'),
         Index(['employment_type',
                'county'], dtype='object')))),
        ('classifier',
         LogisticRegression(class_weight='balanced', max_iter=1000,
                             random_state=42)))]))

```

```

[86]: # Make predictions
      # We'll generate both class predictions and probabilities.
      # Class predictions
      y_pred = logistic_model.predict(x_test)

      # this produces 0 = predicted non-default and 1 = predicted default

[87]: # Default probabilities
      # For example, the model might produce: Borrower A → 0.08, Borrower B → 0.31, B
            ↪ Borrower C → 0.74
      # Meaning the model estimates different probabilities of default.
      y_prob = logistic_model.predict_proba(x_test)[:, 1]

[88]: # Evaluate the model
      # Import the evaluation metrics:

      from sklearn.metrics import (
          accuracy_score,
          precision_score,
          recall_score,
          f1_score,
          confusion_matrix,
          classification_report,
          roc_auc_score
      )

[89]: # Calculate the basic metrics:
      accuracy = accuracy_score(y_test, y_pred)
      precision = precision_score(y_test, y_pred)
      recall = recall_score(y_test, y_pred)
      f1 = f1_score(y_test, y_pred)
      roc_auc = roc_auc_score(y_test, y_prob)

      print("Accuracy:", accuracy)
      print("Precision:", precision)

```

```
print("Recall:", recall)
print("F1 Score:", f1)
print("ROC-AUC:", roc_auc)
```

Accuracy: 0.7179
Precision: 0.20965732087227415
Recall: 0.703239289446186
F1 Score: 0.3230141588672906
ROC-AUC: 0.7808098102286407

```
[90]: print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.96	0.72	0.82	9043
1	0.21	0.70	0.32	957
accuracy			0.72	10000
macro avg	0.58	0.71	0.57	10000
weighted avg	0.89	0.72	0.77	10000

```
[91]: cm = confusion_matrix(y_test, y_pred)
```

```
print(cm)
```

```
[[6506 2537]
 [ 284  673]]
```

```
[97]: # Test different probability thresholds
import pandas as pd
from sklearn.metrics import precision_score, recall_score, f1_score

thresholds = [0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80]

threshold_results = []

for threshold in thresholds:
    y_pred_threshold = (y_prob >= threshold).astype(int)

    threshold_results.append({
        'threshold': threshold,
        'precision': precision_score(y_test, y_pred_threshold),
        'recall': recall_score(y_test, y_pred_threshold),
        'f1_score': f1_score(y_test, y_pred_threshold)
    })

threshold_df = pd.DataFrame(threshold_results)
```

```
threshold_df
```

```
[97]:
```

	threshold	precision	recall	f1_score
0	0.2	0.118622	0.957158	0.211084
1	0.3	0.137752	0.886102	0.238437
2	0.4	0.165734	0.804598	0.274853
3	0.5	0.209657	0.703239	0.323014
4	0.6	0.277442	0.575758	0.374448
5	0.7	0.357731	0.408568	0.381463
6	0.8	0.453125	0.212121	0.288968

```
[98]: # Find the threshold with the highest F1
best_threshold = threshold_df.loc[
    threshold_df['f1_score'].idxmax()
]

best_threshold
```

```
[98]: threshold    0.700000
precision    0.357731
recall       0.408568
f1_score     0.381463
Name: 5, dtype: float64
```

```
[99]: # Visualize the trade-off
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 5))

plt.plot(
    threshold_df['threshold'],
    threshold_df['precision'],
    marker='o',
    label='Precision'
)

plt.plot(
    threshold_df['threshold'],
    threshold_df['recall'],
    marker='o',
    label='Recall'
)

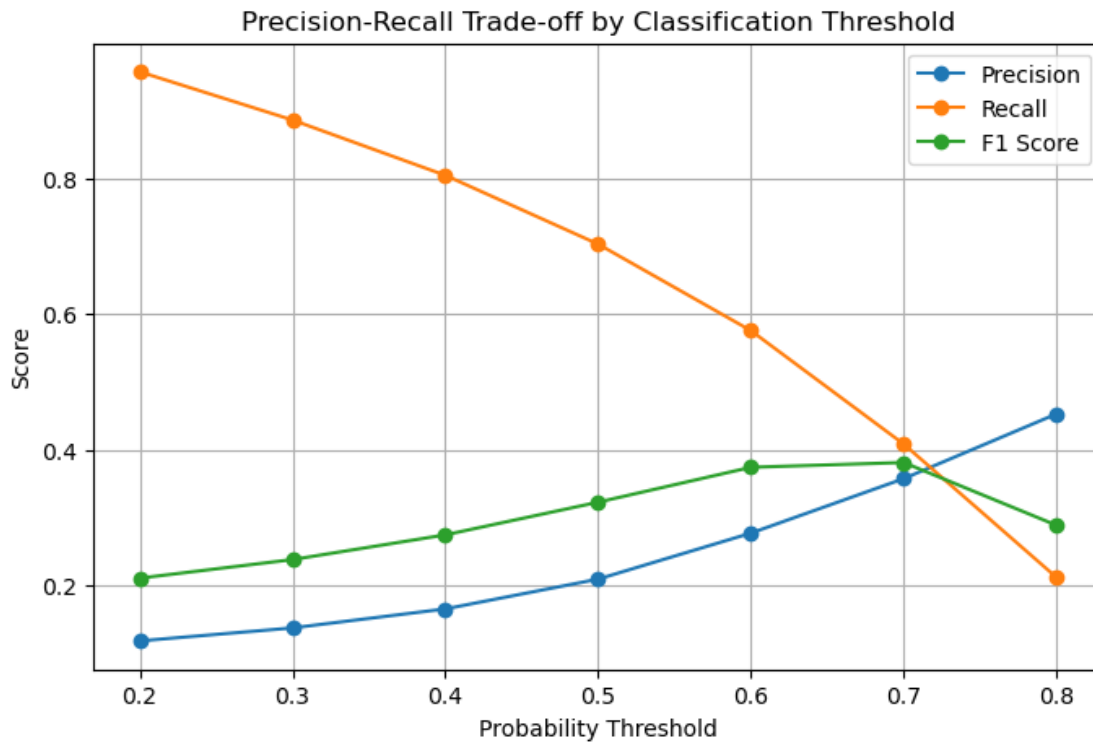
plt.plot(
    threshold_df['threshold'],
    threshold_df['f1_score'],
    marker='o',
```

```

    label='F1 Score'
)

plt.xlabel('Probability Threshold')
plt.ylabel('Score')
plt.title('Precision-Recall Trade-off by Classification Threshold')
plt.legend()
plt.grid()
plt.show()

```



```

[100]: # Gradient Boosting
from sklearn.ensemble import GradientBoostingClassifier

gradient_model = Pipeline(
    steps=[
        ('preprocessor', preprocessor),
        (
            'classifier',
            GradientBoostingClassifier(
                n_estimators=100,
                learning_rate=0.1,
                max_depth=3,
                random_state=42
            )
        )
    ]
)

```

```
)
    )
]
)
```

```
[101]: # Then train
gradient_model.fit(x_train, y_train)
```

```
[101]: Pipeline(steps=[('preprocessor',
                        ColumnTransformer(transformers=[('num', StandardScaler(),
                                                         Index(['age',
                                                         'membership_years', 'monthly_income_kes', 'monthly_cash_flow',
                                                         'savings_to_income_ratio', 'loan_to_income_ratio', 'loan_term_months',
                                                         'monthly_repayment_kes', 'previous_loans', 'previous_defaults',
                                                         'previous_late_payments', 'utility_payment_consistency',
                                                         'income_consistency_score', 'total_debt_service_ratio'],
                                                         dtype='object')),
                                                         ('cat',
                                                         OneHotEncoder(drop='first',
                                                         Index(['employment_type',
                                                         'county'], dtype='object')))])),
                        ('classifier', GradientBoostingClassifier(random_state=42))])
```

```
[102]: # Generate predictions
y_pred_gb = gradient_model.predict(x_test)

y_prob_gb = gradient_model.predict_proba(x_test)[: , 1]
```

```
[103]: # And evaluate
accuracy_gb = accuracy_score(y_test, y_pred_gb)
precision_gb = precision_score(y_test, y_pred_gb)
recall_gb = recall_score(y_test, y_pred_gb)
f1_gb = f1_score(y_test, y_pred_gb)
roc_auc_gb = roc_auc_score(y_test, y_prob_gb)

print("Accuracy:", accuracy_gb)
print("Precision:", precision_gb)
print("Recall:", recall_gb)
print("F1 Score:", f1_gb)
print("ROC-AUC:", roc_auc_gb)
```

```
Accuracy: 0.9192
Precision: 0.7898832684824902
Recall: 0.21212121212121213
F1 Score: 0.3344316309719934
ROC-AUC: 0.8072047159796496
```

```
[104]: cm_gb = confusion_matrix(y_test, y_pred_gb)

print(cm_gb)
```

```
[[8989   54]
 [ 754 203]]
```

```
[105]: # test different thresholds for Gradient Boosting
thresholds = [0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80]

gb_threshold_results = []

for threshold in thresholds:
    y_pred_threshold = (y_prob_gb >= threshold).astype(int)

    gb_threshold_results.append({
        'threshold': threshold,
        'precision': precision_score(y_test, y_pred_threshold),
        'recall': recall_score(y_test, y_pred_threshold),
        'f1_score': f1_score(y_test, y_pred_threshold)
    })

gb_threshold_df = pd.DataFrame(gb_threshold_results)

gb_threshold_df
```

```
[105]:
```

	threshold	precision	recall	f1_score
0	0.1	0.284463	0.644723	0.394754
1	0.2	0.463734	0.407524	0.433815
2	0.3	0.607143	0.301985	0.403350
3	0.4	0.709677	0.252874	0.372881
4	0.5	0.789883	0.212121	0.334432
5	0.6	0.839806	0.180773	0.297506
6	0.7	0.879310	0.159875	0.270557
7	0.8	0.931034	0.141066	0.245009

```
[106]: # highest F1 among those thresholds
best_gb_threshold = gb_threshold_df.loc[
    gb_threshold_df['f1_score'].idxmax()
]

best_gb_threshold
```

```
[106]: threshold    0.200000
precision    0.463734
recall       0.407524
f1_score     0.433815
Name: 1, dtype: float64
```

```
[107]: # tuning the gradient boost model
from sklearn.model_selection import GridSearchCV

param_grid = {
    'classifier__n_estimators': [100, 200],
    'classifier__learning_rate': [0.05, 0.10],
    'classifier__max_depth': [2, 3, 4]
}
```

```
[108]: grid_search = GridSearchCV(
    estimator=gradient_model,
    param_grid=param_grid,
    scoring='roc_auc',
    cv=3,
    n_jobs=-1,
    verbose=1
)
```

```
[109]: # STEP 3 - RUN THE TUNING
grid_search.fit(x_train, y_train)
```

Fitting 3 folds for each of 12 candidates, totalling 36 fits

```
[109]: GridSearchCV(cv=3,
                    estimator=Pipeline(steps=[('preprocessor',
                                                ColumnTransformer(transformers=[('num',
                                                                                      StandardScaler(),
                                                                                      Index(['age', 'membership_years', 'monthly_income_kes', 'monthly_cash_flow',
                                                                                      'savings_to_income_ratio', 'loan_to_income_ratio', 'loan_term_months',
                                                                                      'monthly_repayment_kes', 'previous_loans', 'previous_defaults',
                                                                                      'previous_late_payments', 'utilit...
                                                                                      dtype='object'))],
                                                                                      ('cat',
                                                                                      OneHotEncoder(drop='first',
                                                                                      handle_unknown='ignore')),
                                                                                      Index(['employment_type', 'county'], dtype='object')))])),
                    ('classifier',
                    GradientBoostingClassifier(random_state=42))]),
                    n_jobs=-1,
                    param_grid={'classifier__learning_rate': [0.05, 0.1],
                                'classifier__max_depth': [2, 3, 4],
                                'classifier__n_estimators': [100, 200]},
                    scoring='roc_auc', verbose=1)
```

```
[110]: print("Best parameters:")
print(grid_search.best_params_)
```

Best parameters:

```
{'classifier__learning_rate': 0.1, 'classifier__max_depth': 2,  
'classifier__n_estimators': 200}
```

```
[111]: print("Best cross-validation ROC-AUC:")  
print(grid_search.best_score_)
```

```
Best cross-validation ROC-AUC:  
0.8049631345213769
```

```
[112]: # Get the tuned model  
tuned_gb = grid_search.best_estimator_
```

```
[114]: # generate predictions on the test set  
y_pred_tuned = tuned_gb.predict(x_test)  
  
y_prob_tuned = tuned_gb.predict_proba(x_test)[:, 1]
```

```
[115]: # evaluate  
accuracy_tuned = accuracy_score(y_test, y_pred_tuned)  
precision_tuned = precision_score(y_test, y_pred_tuned)  
recall_tuned = recall_score(y_test, y_pred_tuned)  
f1_tuned = f1_score(y_test, y_pred_tuned)  
roc_auc_tuned = roc_auc_score(y_test, y_prob_tuned)  
  
print("Accuracy:", accuracy_tuned)  
print("Precision:", precision_tuned)  
print("Recall:", recall_tuned)  
print("F1 Score:", f1_tuned)  
print("ROC-AUC:", roc_auc_tuned)
```

```
Accuracy: 0.918  
Precision: 0.7665369649805448  
Recall: 0.2058516196447231  
F1 Score: 0.3245469522240527  
ROC-AUC: 0.8066619706543138
```

```
[116]: # And the confusion matrix:  
cm_tuned = confusion_matrix(y_test, y_pred_tuned)  
  
print(cm_tuned)
```

```
[[8983   60]  
 [ 760  197]]
```

```
[118]: # tuned model threshold analysis  
thresholds = [0.10, 0.15, 0.20, 0.25, 0.30,  
              0.40, 0.50, 0.60, 0.70, 0.80]  
tuned_threshold_results = []
```

```

for threshold in thresholds:
    y_pred_threshold = (y_prob_tuned >= threshold).astype(int)

    tuned_threshold_results.append({
        'threshold': threshold,
        'precision': precision_score(
            y_test,
            y_pred_threshold
        ),
        'recall': recall_score(
            y_test,
            y_pred_threshold
        ),
        'f1_score': f1_score(
            y_test,
            y_pred_threshold
        )
    })

tuned_threshold_df = pd.DataFrame(
    tuned_threshold_results
)
tuned_threshold_df

```

```

[118]: threshold precision recall f1_score
0      0.10   0.275524  0.645768  0.386250
1      0.15   0.374439  0.523511  0.436601
2      0.20   0.447172  0.437827  0.442450
3      0.25   0.514706  0.365726  0.427611
4      0.30   0.587549  0.315569  0.410605
5      0.40   0.679191  0.245559  0.360706
6      0.50   0.766537  0.205852  0.324547
7      0.60   0.848958  0.170324  0.283725
8      0.70   0.890909  0.153605  0.262032
9      0.80   0.938462  0.127482  0.224471

```

```

[119]: best_tuned_threshold = tuned_threshold_df.loc[
        tuned_threshold_df['f1_score'].idxmax()
    ]

best_tuned_threshold

```

```

[119]: threshold    0.200000
precision    0.447172
recall       0.437827
f1_score     0.442450
Name: 2, dtype: float64

```

```
[120]: # Step 1 - Create a validation set from the training data
from sklearn.model_selection import train_test_split

x_train_inner, x_val, y_train_inner, y_val = train_test_split(
    x_train,
    y_train,
    test_size=0.20,
    random_state=42,
    stratify=y_train
)

print("Inner training:", x_train_inner.shape)
print("Validation:", x_val.shape)
```

Inner training: (32000, 16)
Validation: (8000, 16)

```
[121]: # Step 2 - Create the tuned Gradient Boosting model again
final_gb = Pipeline(
    steps=[
        ('preprocessor', preprocessor),
        (
            'classifier',
            GradientBoostingClassifier(
                n_estimators=200,
                learning_rate=0.1,
                max_depth=2,
                random_state=42
            )
        )
    ]
)
```

```
[122]: # Train it on the inner training set:
final_gb.fit(x_train_inner, y_train_inner)
```

```
[122]: Pipeline(steps=[('preprocessor',
                        ColumnTransformer(transformers=[('num', StandardScaler(),
                                                         Index(['age',
                                                         'membership_years', 'monthly_income_kes', 'monthly_cash_flow',
                                                         'savings_to_income_ratio', 'loan_to_income_ratio', 'loan_term_months',
                                                         'monthly_repayment_kes', 'previous_loans', 'previous_defaults',
                                                         'previous_late_payments', 'utility_payment_consistency',
                                                         'income_consistency_score', 'total_debt_service_ratio'],
                                                         dtype='object')),
                        ('cat',
                         OneHotEncoder(drop='first',
                         handle_unknown='ignore'),
```

```

Index(['employment_type',
'county'], dtype='object')))),
      ('classifier',
      GradientBoostingClassifier(max_depth=2, n_estimators=200,
                                random_state=42)))]

```

```

[124]: # Generate validation probabilities:
y_val_prob = final_gb.predict_proba(x_val)[: , 1]

```

```

[125]: # Test a wider range of thresholds:
thresholds = [
    0.05, 0.10, 0.15, 0.20,
    0.25, 0.30, 0.35, 0.40,
    0.45, 0.50, 0.60, 0.70
]

validation_results = []

for threshold in thresholds:

    y_val_pred = (y_val_prob >= threshold).astype(int)

    validation_results.append({
        'threshold': threshold,
        'precision': precision_score(y_val, y_val_pred),
        'recall': recall_score(y_val, y_val_pred),
        'f1_score': f1_score(y_val, y_val_pred)
    })

validation_df = pd.DataFrame(validation_results)

validation_df

```

```

[125]:
threshold  precision  recall  f1_score
0         0.05    0.156558  0.830065  0.263431
1         0.10    0.260377  0.631373  0.368702
2         0.15    0.341615  0.503268  0.406977
3         0.20    0.434159  0.435294  0.434726
4         0.25    0.493805  0.364706  0.419549
5         0.30    0.563927  0.322876  0.410640
6         0.35    0.637143  0.291503  0.400000
7         0.40    0.690141  0.256209  0.373689
8         0.45    0.740741  0.235294  0.357143
9         0.50    0.775120  0.211765  0.332649
10        0.60    0.847682  0.167320  0.279476
11        0.70    0.915966  0.142484  0.246606

```

```
[126]: # best validation threshold
best_validation_threshold = validation_df.loc[
    validation_df['f1_score'].idxmax()
]

best_validation_threshold
```

```
[126]: threshold    0.200000
precision    0.434159
recall       0.435294
f1_score     0.434726
Name: 3, dtype: float64
```

```
[127]: # Retrain the final model
final_gb.fit(x_train, y_train)
```

```
[127]: Pipeline(steps=[('preprocessor',
                        ColumnTransformer(transformers=[('num', StandardScaler(),
                                                         Index(['age',
                                                         'membership_years', 'monthly_income_kes', 'monthly_cash_flow',
                                                         'savings_to_income_ratio', 'loan_to_income_ratio', 'loan_term_months',
                                                         'monthly_repayment_kes', 'previous_loans', 'previous_defaults',
                                                         'previous_late_payments', 'utility_payment_consistency',
                                                         'income_consistency_score', 'total_debt_service_ratio'],
                                                         dtype='object'))),
                        ('cat',
                         OneHotEncoder(drop='first',
                                         Index(['employment_type',
                                         'county'], dtype='object')))],
                        ('classifier',
                         GradientBoostingClassifier(max_depth=2, n_estimators=200,
                                                         random_state=42))])
```

```
[128]: # Generate probabilities for the untouched test set:
y_test_prob_final = final_gb.predict_proba(x_test)[: , 1]
```

```
[129]: # Apply the threshold selected from validation:
final_threshold = best_validation_threshold['threshold']

y_test_pred_final = (
    y_test_prob_final >= final_threshold
).astype(int)
```

```
[130]: # Step 5 - Final evaluation
final_accuracy = accuracy_score(
    y_test,
```

```

        y_test_pred_final
    )

    final_precision = precision_score(
        y_test,
        y_test_pred_final
    )

    final_recall = recall_score(
        y_test,
        y_test_pred_final
    )

    final_f1 = f1_score(
        y_test,
        y_test_pred_final
    )

    final_roc_auc = roc_auc_score(
        y_test,
        y_test_prob_final
    )

    print("Final threshold:", final_threshold)
    print("Accuracy:", final_accuracy)
    print("Precision:", final_precision)
    print("Recall:", final_recall)
    print("F1 Score:", final_f1)
    print("ROC-AUC:", final_roc_auc)

```

```

Final threshold: 0.2
Accuracy: 0.8944
Precision: 0.4471718249733191
Recall: 0.4378265412748171
F1 Score: 0.4424498416050686
ROC-AUC: 0.8066619706543138

```

```

[131]: # and the confusion matrix
final_cm = confusion_matrix(
    y_test,
    y_test_pred_final
)

print(final_cm)

```

```

[[8525  518]
 [ 538  419]]

```

```
[137]: # build the credit-risk scoring system
credit_scores = x_test.copy()

credit_scores['actual_default'] = y_test.values

credit_scores['default_probability'] = y_test_prob_final

credit_scores['risk_score'] = (
    y_test_prob_final * 100
)
```

```
[138]: # Create risk bands
credit_scores['risk_band'] = pd.cut(
    credit_scores['default_probability'],
    bins=[0, 0.10, 0.20, 1.00],
    labels=[
        'Low Risk',
        'Moderate Risk',
        'Higher Risk'
    ],
    include_lowest=True
)
```

```
[139]: # See the resulting borrowers
credit_scores[
    [
        'default_probability',
        'risk_score',
        'risk_band',
        'actual_default'
    ]
].head(20)
```

	default_probability	risk_score	risk_band	actual_default
44284	0.019664	1.966413	Low Risk	0
41181	0.125422	12.542157	Moderate Risk	0
6066	0.285688	28.568808	Higher Risk	1
21869	0.326630	32.663037	Higher Risk	0
29239	0.095072	9.507205	Low Risk	0
29079	0.026808	2.680761	Low Risk	0
42451	0.040424	4.042438	Low Risk	0
3000	0.051584	5.158430	Low Risk	0
31180	0.040842	4.084215	Low Risk	0
32911	0.055013	5.501345	Low Risk	0
49986	0.021590	2.158950	Low Risk	0
41016	0.879875	87.987549	Higher Risk	1
23766	0.239105	23.910488	Higher Risk	0

43449	0.009616	0.961578	Low Risk	0
38127	0.016607	1.660669	Low Risk	0
40891	0.048050	4.804953	Low Risk	1
41518	0.124276	12.427563	Moderate Risk	0
10827	0.288279	28.827880	Higher Risk	1
11674	0.086745	8.674471	Low Risk	0
30822	0.058999	5.899897	Low Risk	0

```
[140]: # Analyze the risk bands
risk_summary = (
    credit_scores
    .groupby('risk_band', observed=True)
    .agg(
        borrowers=('default_probability', 'count'),
        average_probability=('default_probability', 'mean'),
        actual_defaults=('actual_default', 'sum'),
        actual_default_rate=('actual_default', 'mean')
    )
)

risk_summary
```

```
[140]:
```

	borrowers	average_probability	actual_defaults \
risk_band			
Low Risk	7757	0.044754	339
Moderate Risk	1306	0.137204	199
Higher Risk	937	0.427195	419

	actual_default_rate
risk_band	
Low Risk	0.043702
Moderate Risk	0.152374
Higher Risk	0.447172

```
[ ]:
```